Benign Prostatic Hyperplasia Rapid Revision

Campbell-Walsh-Wein · Chapters 148–152

For MCh Urology theory & viva · NEET-SS / competitive MCQs

How to use (total ≈ 3h 52min even at a slow, careful pace)

- Each chapter reads top-to-bottom in the order you would write an essay answer: definition → pathology → diagnosis → management → complications → follow-up. The same material serves theory and viva.
- Boxed **ALGORITHM** and shaded **KEY POINT** panels carry the reasoning examiners ask about ("why do we…?").
- **MCQ one-liners** end each chapter: re-read these the day before the paper.
- The last section puts the big treatment choices side by side for essay questions.

Conventions: **underlined bold** = number to memorise · MCQ = exam trivia only · → *Ch 151* = cross-reference · † = not in the source chapter notes (added because it is essential in practice). Abbreviations are spelled out at first use in each chapter.

Chapter	Topic	Slow-reader time
148	Anatomy & Development of the Prostate	15 min
149	BPH: Aetiology, Pathophysiology & Natural History	62 min
150	Evaluation & Nonsurgical Management of BPH	73 min
151	Minimally Invasive & Endoscopic Management of BPH	58 min
152	Simple Prostatectomy	19 min
★	Putting it together: the BPH pathway	5 min
	Total (16271 words, at 70 words/min)	**3h 52min**

148 Anatomy and Development of the Prostate

Reading time ≈ 15 min (slow pace)

1 · DEVELOPMENT (EMBRYOLOGY)

- **Origin**: urogenital sinus (UGS) — transient urethral widening near the bladder neck, formed by cloacal division; endoderm-derived epithelial tube in a mesoderm-derived mesenchymal sheath. Initially ambisexual: female UGS also forms prostate if given enough androgen.
- Incomplete cloacal division (imperforate anus, urorectal fistula) → impaired prostate development.
- **Androgen/DHT**: †testosterone is converted in UGS mesenchyme by 5α-reductase type 2 to dihydrotestosterone (DHT), the more potent androgen receptor (AR) ligand. Loss-of-function **SRD5A2** variant → prostate incomplete; castration or inactive AR (rodents) → prostate absent. †Same enzyme is the target of 5α-reductase inhibitors (5-ARI) in benign prostatic hyperplasia (BPH).
- **Androgen is not rate-limiting**: testosterone present by week 9, budding only at week 10. The gate is **AR demethylation** in the UGS (epigenetic checkpoint) — receptor availability, not hormone supply.

Week	Event
6–7	Cloaca divides → ventral UGS, dorsal anorectum
≈9	Testosterone activates **mesenchymal** AR → andromedins instruct epithelium
10	Solid epithelial buds invade mesenchyme; bud pattern fixes adult ductal pattern
11 →	Elongation, canalisation (urethra → bud tips), branching through puberty

Mesenchyme instructs, epithelium responds

- **Lasnitzki–Mizuno** tissue recombination: buds form only when **mesenchyme** carries functional AR — epithelial AR not needed for initiation.
- **Cunha** recombination: UGS mesenchyme reprogrammed adult bladder epithelium into prostatic glands → **instructive** (not permissive) induction.
- Epithelial specification: coexpression of **Nkx3-1**, Edar, Wnt10b.
- **Andromedins**: androgen-induced stromal signals (fibroblast growth factor 10 [FGF10] — required for development; FGF7, Wif1, Wnt2). No single factor proven to meet all three criteria; FGF10 and retinoic acid have major roles.
- Branching also needs **stromal** AR. Sonic hedgehog is the epithelial factor acting on mesenchyme (paracrine). β-catenin sets bud size and spacing.
- Oestrogens: not needed for budding; may pattern branching.

> **KEY POINT** McNeal: UGS mesenchyme loses inductive capacity with age; **re-awakening of embryonic induction** in ageing TZ stroma may initiate new glands → BPH.

- **Environmental** (developmental origins of health and disease): gestational diethylstilboestrol → squamous metaplasia in human prostate; phthalates, dioxin (tetrachlorodibenzo-p-dioxin, TCDD), vinclozolin → agenesis or transgenerational epimutations. TCDD → **hyperinnervation** → hyperactive smooth muscle, frequency — one explanation for variable α-blocker response.

2 · ZONAL ANATOMY (MCNEAL)

- Base = bladder neck; apex = junction with membranous urethra. Prostatic urethra bends **≈35°** at the verumontanum; **13–15** ducts open dorsolaterally.
- **Verumontanum**: bulge distal to the urethral crest where ejaculatory ducts enter — distal limit of transurethral resection of prostate (TURP)/enucleation (sphincter just beyond).

Zone	Ducts	†% glandular tissue	Disease
Transition (TZ)	Shortest; glands closest to proximal urethra	≈5	**BPH**; †≈20% of cancers
Peripheral (PZ)	Longest, distal	≈70	**70%** of cancers; †palpable on digital rectal examination
Central (CZ)	Surround ejaculatory ducts (base)	≈25	†Few cancers
Anterior fibromuscular stroma	None — fibroblasts + smooth muscle	Non-glandular	—

KEY POINT Trap: CZ (around ejaculatory ducts) ≠ TZ. Short proximal ducts **enlarge** (BPH); long distal ducts **transform** (cancer). Enlarging TZ compresses PZ into the **surgical (pseudo)capsule** — the enucleation plane → *Ch 152*.

3 · FASCIA, NEUROVASCULAR BUNDLES, VESSELS, LYMPHATICS †

- †**Capsule**: fibromuscular, incomplete anteriorly/apically. **Endopelvic fascia** laterally; **Denonvilliers fascia** posteriorly separates prostate/seminal vesicles from rectum (plane for radical prostatectomy). Puboprostatic ligaments anchor anteriorly.
- †**Neurovascular bundles**: posterolateral (5 and 7 o'clock), between lateral pelvic and prostatic fascia; carry cavernous nerves from pelvic plexus → erection. Lie outside the capsule — hence spared by enucleation.
- †**Arteries**: inferior vesical/prostatic artery (internal iliac) → urethral branches (enter at bladder neck, 5 and 7 o'clock, supply TZ/adenoma) and capsular branches.
- †**Veins**: dorsal venous complex (Santorini plexus) under the pubis → internal iliac; also communicates with vertebral (Batson) plexus — route for spinal metastases.
- †**Lymphatics**: obturator, internal and external iliac, presacral nodes.

4 · HISTOLOGY AND CELL TYPES

- Acinus: pseudostratified epithelium (basal + luminal) on basement membrane, wrapped by periepithelial fibroblasts; interacinar fibroblasts + smooth muscle.

Feature	Luminal (secretory)	Basal
Keratins	8, 18	5, 14
Marker	PSA, NKX3-1	KRT5/14
AR	High	Low
Secretes	PSA, prostatic acid phosphatase (PAP)	Almost none
Role	Barrier, secretion; terminally differentiated	Ductal integrity; can be luminal progenitor

- **Urethra + proximal ducts**: lung-like epithelium — **club** (SCGB1A1), **hillock** (KRT13), neuroendocrine (CHGA; mostly neural crest–derived), ionocytes. Handover to NKX3-1+ epithelium = cellular definition of the "transition zone".
- BPH stroma: prominent T, B and dendritic cells — resembles mucosal inflammation.

5 · SECRETIONS AND CLINICAL MARKERS

- Prostatic fluid: **37–44%** of semen, acidic (pH ≈6.3); seminal vesicle fluid (majority, alkaline) supplies semenogelins.
- Coagulation → liquefaction by kallikreins within **20–30 min**; nonliquefaction if incomplete by 60 min.

- **Ion trapping**: drugs concentrate where most ionised — macrolides in acidic prostatic fluid; inflammation alkalinises fluid.
- "Prostate-specific" is a misnomer: PSA, PSMA and PAP are all extraprostatic.

Product	Key facts
Prostate-specific antigen (PSA; kallikrein 3, KLK3)	Serine protease, 19q13.4 kallikrein cluster; liquefies semen; serum biomarker
Prostate-specific membrane antigen (PSMA)	Folate hydrolase; rarely deleted in cancer → PSMA-PET, ^{177}Lu-PSMA therapy
PAP	First tumour marker MCQ (Huggins and Hodges, 1941)
Zinc	Highest of any tissue; inhibits aconitase → citrate secreted

KEY POINT Benign cells: high zinc blocks aconitase → citrate exported (inefficient metabolism). Cancer: **zinc lost**, citrate oxidised → low citrate (†basis of magnetic resonance spectroscopy).

6 · MCQ ONE-LINERS

1. Prostate derives from urogenital sinus (endoderm).
2. Budding begins at week 10.
3. Initiation requires AR in mesenchyme, not epithelium.
4. Gate to budding: AR gene demethylation.
5. UGS mesenchyme is an instructive inducer (Cunha).
6. SRD5A2 deficiency → incomplete prostate.
7. Andromedin required for development: FGF10.
8. Epithelial specification marker: Nkx3-1.
9. BPH arises in transition zone.
0. 70% of cancers arise in peripheral zone.
1. Central zone surrounds ejaculatory ducts.
2. Anterior fibromuscular stroma is non-glandular.
3. Prostatic urethra angulates ≈35° at verumontanum.
4. Luminal keratins 8/18; basal 5/14.
15. Hillock cell marker: KRT13; club: SCGB1A1.
16. Most neuroendocrine cells are neural crest–derived.
17. Mouse has no PSA gene.
18. Liquefaction normal within 20–30 minutes.
19. Highest tissue zinc: prostate.
20. Zinc inhibits mitochondrial aconitase.
21. Cancer: low zinc, low citrate.
22. PSMA is a folate hydrolase.
23. PAP: first prostate cancer marker.
24. Gestational DES → squamous metaplasia.
25. TCDD → prostatic hyperinnervation.
26. †Neurovascular bundles: posterolateral, outside capsule.
27. †Adenoma arterial supply enters at 5 and 7 o'clock.

149 Benign Prostatic Hyperplasia: Aetiology, Pathophysiology, Epidemiology and Natural History

Reading time ≈ 62 min (slow pace)

1 · DEFINITIONS AND THE FOUR OVERLAPPING ENTITIES

Term	What it is	How measured
Benign prostatic hyperplasia (BPH)	Histological process: ↑ number of epithelial and stromal cells in the periurethral / transition zone (TZ)	Histology (surgical or autopsy)
Benign prostatic enlargement (BPE)	Gland enlargement resulting from histological BPH	Digital rectal examination (DRE), transrectal ultrasonography (TRUS), magnetic resonance imaging (MRI)
Bladder outlet obstruction (BOO)	Obstruction of outflow from any cause (BPH, stricture, cancer, bladder neck sclerosis)	Pressure–flow study (best); flow rate (indirect)
Benign prostatic obstruction (BPO)	BOO caused by BPE	Pressure–flow study
Lower urinary tract symptoms (LUTS)	Symptom complex: storage, voiding, post-micturition; not prostate-specific	International Prostate Symptom Score (IPSS)

- The four circles (histological BPH, BPE, BOO, bothersome LUTS) overlap only **partially**. BPE without LUTS and LUTS without BPE are both common; BOO may exist without LUTS, without BPE, or without BPH at all (e.g. stricture).
- LUTS are often, but not invariably, due to BOO from BPO, which in turn often accompanies BPE from histological BPH.
- **Non-prostatic causes of male LUTS**: age-related detrusor dysfunction, polyuria, sleep disorders, neurogenic and primary bladder disease. Impaired detrusor contractility mimics obstruction with no obstruction at all; voiding symptoms correlate poorly with the underlying pathophysiology.
- "**Prostatism**" is obsolete and wrong: it made the prostate the sole source of LUTS. Paul Abrams coined "LUTS" to replace it MCQ.
- Storage symptoms largely = **overactive bladder (OAB)** (urgency, frequency, nocturia, urgency incontinence; reflects detrusor overactivity, DO). Storage symptoms bother more than voiding symptoms, especially with incontinence. Post-micturition dribble is commonest in men and highly troublesome. Age-matched women report similar LUTS frequency and severity.
- "Hypertrophy" is pathologically incorrect: BPH is a net ↑ in cell **number**, not cell size.
- † Urodynamic BOO is graded by the bladder outlet obstruction index (BOOI = PdetQmax − 2Qmax): **>40 obstructed, 20–40 equivocal, <20 unobstructed**.

> **KEY POINT** Viva: "Is every man with LUTS and a big prostate obstructed?" No. Size, symptoms, flow and obstruction are four separate dimensions; obstruction can only be diagnosed by measuring it (pressure–flow).

2 · EPIDEMIOLOGY (HEADLINE)

Histological prevalence

- Pooled autopsy data: **no BPH under 30**; prevalence rises from the fourth decade, ≈ **50% in the 50s**†, peaking at **88% in the 80s** (approaches 100% in the ninth decade).
- Age-specific autopsy prevalence is **strikingly similar across all ethnic and geographic populations** (including India, China, Japan).
- Histological prevalence is comparable across studies; clinical prevalence depends entirely on the definition used. No globally accepted epidemiological definition of BPH exists. For a benign

disease, **prevalence** matters more than incidence.

Clinical classification — three independent axes

- **Symptoms** (IPSS), **enlargement** (DRE/TRUS/MRI), **obstruction** (flow or pressure–flow). Plus bother, interference with activities and quality of life (QoL). "Clinical BPH" is an imprecise term for this constellation.
- Proposed cut-offs: volume **>30 mL**; common community composite: IPSS >7, maximum flow rate (Qmax) <15 mL/s, volume >20 mL.

IPSS band	Score
Mild	**0–7**
Moderate	**8–19**
Severe	**20–35**

- IPSS = American Urological Association (AUA) Symptom Index (seven items, 0–35) + QoL question; self-administered, validated across languages, robust to socioeconomic status and mode of administration.
- **Shortcoming**: no items on incontinence or pain. The Lower Urinary Tract Research Network (LURN) questionnaire fills this gap.
- General QoL (Short-Form 36, SF-36) is only marginally affected by LUTS → disease-specific bother/QoL instruments are needed.

Size and flow with age

Method	Performance
DRE	Poor inter-observer reliability; **underestimates** size by 25% to ≥50% as glands enlarge
TRUS (prolate ellipsoid)	Most widely accepted; 3D TRUS more accurate
MRI	Research standard with TRUS; reads ≈10% larger than TRUS
Computed tomography	Less accurate; adequate clinically

- Total prostate volume (TPV) ≈ **25 mL in the 30s → 35–45 mL in the 70s**; transition zone volume (TZV) ≈ 15 → 25 mL.
- Qmax falls almost linearly with age (Olmsted County median 20.3 mL/s at 40–44 → 11.3 mL/s at 75–79).

Qmax	Probability of obstruction
<10 mL/s	High
10–15 mL/s	Intermediate
>15 mL/s	Low

- Limitations: Qmax depends on voided volume, and has high diurnal and day-to-day variability. Pressure–flow is the gold standard; non-invasive penile cuff testing approximates it.

3 · AETIOLOGY

> **KEY POINT** Core concept: BPH is an **accumulation** disease. Volume = proliferation − death. Established BPH shows equal or reduced replication with ↑ anti-apoptotic genes (bcl-2); androgens act chiefly by **suppressing cell death**. That is why androgen withdrawal shrinks the gland. Early BPH may involve rapid proliferation.

- New gland formation (budding → branching → elongation) otherwise occurs only in fetal life → concept of **embryonic reawakening** of the inductive potential of the stroma.

- Candidate contributors (singly or combined): androgens, oestrogens, stromal–epithelial interaction, growth factors, inflammation, neurotransmitters, genetics.

3.1 Androgens — permissive, not causal

- Men castrated before puberty or with defects of androgen production/action **never develop BPH**. Androgen withdrawal causes **partial** involution of established BPH.
- Two established determinants of BPH: **age** and **functioning testes**.
- Serum testosterone falls with age, yet **intraprostatic dihydrotestosterone (DHT) and androgen receptor (AR) stay high**. Intraprostatic DHT is **maintained, not elevated**, in BPH (earlier "raised DHT" reports used cadaveric controls in whom DHT falls after death) [MCQ].
- Serum testosterone does not predict prostate volume or prostate-specific antigen (PSA). The only variable tracking low testosterone is body mass index (BMI): obesity lowers testosterone. Exception: higher baseline serum DHT predicted higher 20-year BPH risk.
- Prostate keeps AR for life; penile AR falls to negligible levels after puberty, so the penis stops growing despite high androgen.
- Neither testosterone nor DHT is a direct mitogen: they act via autocrine/paracrine growth factors.
- Androgens are also **anti-inflammatory**: AR knockout in mouse epithelium ↑ inflammatory cytokines; hypogonadism is associated with inflammation.
- AR CAG-repeat and vitamin D receptor polymorphisms: inconsistent, at most minor.

ALGORITHM
Testosterone → enters stromal cell → **type 2 5α-reductase** (nuclear membrane, interstitial fibroblast) → DHT → acts on stroma (autocrine) or diffuses to epithelium (paracrine) → DHT–AR complex → DNA → transcription of growth factors → epithelial survival/growth

- ≈ **90%** of prostatic androgen is DHT of testicular origin; ≈10% adrenal (negligible role).
- DHT > testosterone in potency: higher AR affinity and more stable complex.
- Androgen withdrawal: switches off androgen-dependent genes (e.g. PSA), activates programmed cell death genes and inflammatory genes.

Feature	5α-reductase type 1	5α-reductase type 2
Main site	Skin, liver	Prostate: **stromal interstitial fibroblasts**, some basal cells (not luminal epithelium)
5α-reductase deficiency syndrome	Normal	**Mutated enzyme**
Finasteride	Not substantially inhibited	Inhibited
Dutasteride	Inhibited	Inhibited
Role	Possibly prostate cancer	Prostate development and hyperplastic growth

- Finasteride (type 2 only) shrinks the gland as much as dual inhibitors, and roughly as much as castration → type 1 DHT is unimportant in BPH; any extra benefit of dual inhibition comes from blocking peripheral DHT.
- SRD5A2 gene on **chromosome 2p23** [MCQ]: V89L ↓ activity, A49T ↑ activity, long TA repeats ↓ activity; links to BPH inconsistent. Promoter methylation in up to 30% of BPH specimens.

3.2 5α-reductase inhibitor (5-ARI) resistance

- 5-ARIs are the **only medical therapy that reduces prostate volume**.
- Clinical resistance is **not** failure to lower intraprostatic DHT. Likely causes: focal absence of luminal atrophy within glandular nodules (persistent anatomical obstruction), or widespread stromal hyperplasia.
- Luminal survival mechanisms (all linked to inflammation): club-cell transition with high nuclear factor-κB (NF-κB) activity (NF-κB can induce androgen-independent AR variants); epigenetic

silencing of SRD5A2 switching survival to oestrogenic pathways. Inflammation correlates with 5-ARI resistance.
- Proposed definition of resistance: volumetric (volume unchanged) + histological (no atrophy) + biochemical (DHT decreased).
- On 5-ARI or androgen deprivation, human luminal cells undergo lineage plasticity to resemble urethral club cells; in mouse, any surviving luminal cell can regenerate the gland (no dedicated progenitor niche).

3.3 Oestrogens — clear in animals, uncertain in man

- Dog: androgen + oestradiol produces experimental BPH; oestrogen induces AR and "sensitises" the ageing gland to androgen, stimulates stroma and ↑ collagen. Gland size rises while DNA synthesis falls → both hormones inhibit cell death.
- Serum oestrogen rises with age, absolutely or relative to testosterone; larger glands go with higher serum oestradiol. Adipose tissue is the main site of aromatisation.

Property	ERα	ERβ
Site	**Stroma**	**Epithelium**
Action	Stromal proliferation; ↑ fibroblast growth factor (FGF)-2, FGF-7	Antiproliferative, prodifferentiation, anti-inflammatory, antioxidant

- Trap: α = stroma, β = epithelium. An ERβ agonist failed in a randomised placebo-controlled LUTS trial.

3.4 Programmed cell death

- Apoptosis: condensation → fragmentation → phagocytosis by neighbours → lysosomal degradation; no immune activation; needs RNA and protein synthesis (active process).
- After castration, death ↑ in luminal epithelium and distal ducts; transforming growth factor-β (TGF-β) is the likely local mediator.

3.5 Stromal–epithelial interaction

- Canine epithelium grown on plastic dedifferentiates and grows fast; on prostatic collagen it stays differentiated and grows slowly → extracellular matrix (ECM) regulates epithelial phenotype.
- Hypothesis: a **stromal defect removes a normal brake** on epithelial proliferation (and expands stroma autocrinely). Embryonic mesenchyme dictates urogenital sinus epithelial differentiation; BPH = reawakening of this stromal induction.
- Interstitial fibroblasts (the periurethral collar carrying type 2 5α-reductase) are ↑ in BPH, expand the TZ, may induce new branching; **no current drug targets them**.
- CXC chemokines and interleukin (IL)-8 convert fibroblasts → myofibroblasts → collagen, fibrosis → LUT dysfunction.

3.6 Growth factors

Androgen supplies the permission; growth factors (likelier site of the primary defect) supply the signal. First prostatic growth factor identified: basic FGF (FGF-2) [MCQ].

Factor	Source → target	Role in BPH
FGF-2 (bFGF)	Stroma → stroma (autocrine)	↑ 2–3-fold; induced by TGF-β and hypoxia
FGF-7 (keratinocyte growth factor)	Stroma → receptor **only on epithelium** (paracrine)	Leading candidate stromal mediator of hormonal control of epithelium
FGF-10	Stromal smooth muscle → epithelium	↑ by androgens
TGF-β1	Stroma	**Inhibits epithelium** but is **mitogenic for fibroblasts**; induces apoptosis, ECM; BPH stroma shows blunted TGF-β inhibition

Factor	Source → target	Role in BPH
Insulin-like growth factors (IGF-1, -2)	—	Transgenic IGF-1 → enlargement; serum IGF-1 unrelated to volume
Others	Epidermal growth factor (EGF), vascular endothelial growth factor (VEGF), FGF-17	

- **Hypoxia hypothesis**: local hypoxia (e.g. atherosclerosis) → hypoxia-inducible factor 1 → stromal FGF-2/FGF-7 → nodule growth → further hypoxia (vicious cycle). Vascular disease (diabetes) may be a BPH risk factor.
- α-Adrenergic signalling may also modulate growth (α-blockade induces apoptosis in some models).

3.7 Inflammation

- BPH tissue is infiltrated by **activated T cells** secreting VEGF, heparin-binding EGF and FGF-2; cytokines IL-2, IL-7, interferon-γ stimulate stromal proliferation; senescent epithelium secretes IL-8 that drives neighbouring cells.
- Chronic inflammation → focal epithelial cyclo-oxygenase-2 (COX-2) up-regulation.
- BPH as an immune-inflammatory disease: antigens may be prostatic (PSA, prostatic acid phosphatase) or infectious; fibroblasts are the central communicators with the infiltrate. Autoimmune disease ↑ BPH prevalence; tumour necrosis factor (TNF) antagonists associated with ↓ BPH incidence.
- **Inflammation correlates with enlargement and 5-ARI resistance**; causality unproven.

3.8 Metabolic syndrome, obesity and lifestyle

- Metabolic syndrome (obesity, glucose intolerance, dyslipidaemia, hypertension) may ↑ prostatic inflammation → LUTS/BPH.
- Mechanisms: adipose aromatisation (↑ oestrogen, ↓ testosterone); **hyperinsulinaemia → sympathetic overactivity**; low high-density lipoprotein (HDL) cholesterol.
- **Waist circumference** (central adiposity) predicts prostatectomy and frequent LUTS independently of BMI. BMI >35 vs <25: odds ratio **3.5** for prostate ≥40 mL. Biases: DRE under-detects enlargement in the obese; obese men steered away from surgery.
- Non-alcoholic fatty liver disease is associated with BPH.

Modifiable factor	Direction
Diabetes, fasting glucose >110 mg/dL, obesity, central adiposity, hypertension (for LUTS)	↑ risk
Physical activity (any level); moderate alcohol	↓ risk (alcohol ↓ BPH but **not LUTS**; confounded by reluctance to operate on drinkers)
Smoking, diet, vasectomy, sexual activity, rural residence	No consistent effect (**vasectomy is not a risk factor**)
Jewish religion	↑ prostatectomy rate but not BPH diagnosis (health-care access) MCQ

- **Depression** is strongly associated with LUTS (neuroinflammation). LUTS severity correlates strongly with erectile and ejaculatory dysfunction; shared ischaemia is plausible.
- **Drugs worsening LUTS**: α-sympathomimetic cold remedies (↑ outlet tone); daily antidepressants, antihistamines, bronchodilators (+2–3 IPSS points); anticholinergic/adrenergic drugs precipitate retention.
- Physical activity is associated with ↓ BPH/LUTS, but randomised evidence that exercise improves IPSS is very low quality.

3.9 Genetics

- Case–control study of men with large resected glands (>37 g) at young age: first-degree male relatives had ≈ **4-fold** risk of surgically treated BPH (brothers highest, ≈6–8-fold).

- Segregation analysis: **autosomal dominant**. ≈ **50%** of men operated on before 60 may have heritable disease vs ≈9% after 60.
- **Familial BPH** = early onset + **large gland**, not severe symptoms; normal serum androgens and normal 5-ARI response.
- BPH and prostate cancer genetic risks are highly correlated; no single causal mutation identified.

3.10 Other factors

- Only **dogs and humans** develop BPH, although all mammals have the same androgen/growth factor machinery; seminal vesicle never becomes hyperplastic → species-specific cofactors.
- Testicular non-androgenic factors may contribute: intact animals grow more prostate than castrates given the same androgen.

4 · PATHOLOGY

4.1 Anatomy (McNeal zones)

- BPH begins in the **periurethral transition zone**. **Every BPH nodule arises in the TZ or the periurethral region**.
- **TZ**: two separate glands just outside the preprostatic sphincter; ducts arise laterally at the point of urethral angulation near the verumontanum.
- **Periurethral zone**: proximal to TZ duct origins, inside the preprostatic sphincter, glands parallel to the urethra.
- TZ also enlarges with age independently of nodule formation.
- **Middle lobe** must be **periurethral** in origin (no TZ tissue at the bladder neck) MCQ.
- Obstructing patterns: isolated middle lobe; isolated lateral lobes; lateral + middle lobes (trilobar); posterior commissural hyperplasia (median bar).

> KEY POINT **Why does the capsule matter?** The capsule (unique to the human prostate) transmits the pressure of tissue expansion inward onto the urethra. The dog gets BPH but rarely obstruction because it has no capsule. Clinical proof: transurethral incision of the prostate (TUIP) incises the capsule and relieves obstruction with no change in volume. Hence **size does not correlate with obstruction**; geometry, capsule and tone matter more.

4.2 Histology — nodules

Feature	Periurethral (stromal) nodules	TZ (glandular) nodules
Composition	Purely stromal (early)	Glandular proliferation, relatively less stroma
Appearance	Like **embryonic mesenchyme**: pale ground substance, little collagen	Tightly packed glands, taller epithelium
Origin	Interstitial fibroblasts	New duct branches budding from existing ducts → entirely new ductal system

- **Two phases**: first ≈20 years → ↑ nodule number, slow growth; phase 2 → growth of individual (mainly glandular) nodules dominates, producing large nodules.
- Small resected glands: mainly fibromuscular stroma; large (enucleated) glands: mainly epithelial nodules. Stromal:epithelial ratio is pleomorphic (contributes to variable drug response).

4.3 Prostatic smooth muscle

In symptomatic BPH, smooth muscle + connective tissue area density exceeds that of epithelium + lumen. Smooth muscle is the target of α-blockers; epithelium is the main target of 5-ARIs.

5 · PATHOPHYSIOLOGY OF OBSTRUCTION

ALGORITHM
Prostatic hyperplasia → ↑ urethral resistance (static + dynamic) → detrusor compensation (↑ pressure to maintain flow) → cost paid in storage function → storage LUTS ± impaired emptying → (minority) decompensation

- LUTS arise from interacting inputs: BOO (BPH or non-BPH), the detrusor response to obstruction, detrusor ageing, neurogenic disease, primary bladder disease and polyuria.

5.1 Static vs dynamic components

Feature	Dynamic (active) tone	Static (mechanical / passive)
Source	Adrenergic stimulation of prostatic smooth muscle	Tissue bulk compressed by the capsule; series elastic elements of cells and, most importantly, ECM
Receptor	**α1a-adrenoceptor** (most abundant subtype)	Independent of active contraction
Treated by	α-blockers	5-ARI (epithelial shrinkage), surgery; passive tone **not reduced** by α-blockade

- BPH smooth muscle is **dedifferentiated** (↓ smooth muscle myosin heavy chain, ↑ non-muscle isoform); α-blockade further alters contractile gene expression → adrenergic signalling regulates the cell, not just its contraction.
- Stroma turns over slowly: a fall in stromal cell number on androgen ablation may take ≥1 year, so "stromal resistance" is unproven.
- Other contractile regulators: endothelin, kallikrein–kinin, **phosphodiesterase (PDE) 4 and 5** (present in prostate and detrusor → PDE5 inhibitors benefit LUTS/BPH in placebo-controlled trials).
- Autonomic overactivity correlates with symptom score; post-tilt noradrenaline rise predicted TZ size.

5.2 Bladder response to obstruction

- Response is largely adaptive, but **many LUTS arise from obstruction-induced bladder change**, not the obstruction itself. ≈ **One-third** keep significant voiding dysfunction (mostly storage) after surgical relief of obstruction.

	Storage-type change	Emptying-type change
Change	Detrusor overactivity (instability), ↓ compliance	↓ Detrusor contractility
Symptoms	Frequency, urgency, nocturia, urge incontinence	Weak stream, hesitancy, intermittency, ↑ residual
End stage	—	Detrusor failure (minority)

ALGORITHM
Three-stage remodelling: smooth muscle hypertrophy (adaptive) → compensation → decompensation (detrusor underactivity, underactive bladder)

- Stimuli: cyclic stretch, ↑ hydrostatic pressure, **hypoxia/ischaemia**; all layers involved (urothelium, suburothelium, detrusor, ECM, nerves).
- Cellular changes: myosin isoform switch, reversion to a **secretory phenotype** producing ECM, mitochondrial dysfunction, calcium-signalling abnormality, impaired cell-to-cell communication. Unlike skeletal muscle, detrusor adapts badly.
- **Trabeculation** (main endoscopic change) = ↑ detrusor **collagen**; severe trabeculation goes with significant residual urine and advanced disease. No accepted grading system.
- Ageing alone reproduces some of these changes (reduced contractility, altered sensation) independent of obstruction.
- Classic decompensation sequence: normal mucosa → trabeculation → cellules → diverticula → detrusor failure.

KEY POINT Why persisting storage LUTS after transurethral resection of the prostate (TURP)? Obstruction leaves a collagen-rich, dedifferentiated, overactive detrusor that removing the adenoma cannot reverse. Counsel patients with DO preoperatively. Acute urinary retention (AUR) is **not** the inevitable end of this sequence: many men in AUR have adequate detrusor function and a precipitant.

6 · NATURAL HISTORY

6.1 Methods and biases

- Designs: watchful waiting cohorts, control (placebo/sham) arms of trials, longitudinal population studies (best: no selection, no label).
- **Regression to the mean**: trial entry thresholds (high score, low flow) select men at a temporary extreme; values drift back at follow-up → spurious "improvement" with no treatment. The stricter the threshold, the larger the apparent improvement. Part of every placebo effect is mathematical.
- **Restriction of range**: entry criteria truncate values, so in trial populations IPSS, Qmax and volume barely correlate (IPSS–TPV essentially none). Only TPV–TZV (strong) and PSA–volume (moderate) survive.
- Outcomes: **direct** (noticed by patient: symptoms, AUR, surgery, infection, death) vs **indirect/proxy** (Qmax, residual, volume, urodynamics).

6.2 Untreated course

- ≈ **40%** improve symptomatically without active treatment, whether on watchful waiting or placebo (placebo does not beat watching). Placebo response rises with baseline severity; active drug gives ≈ twice the improvement in each stratum.
- Mild LUTS (IPSS <8): ≈ 1 in 3 progress to moderate/severe over 4 years.
- Community: symptoms fluctuate in both directions; ≈1 in 5 with severe symptoms are asymptomatic a year later; regression is as common as progression.
- **Olmsted County** (most informative natural history study):
  - IPSS rises ≈ **0.2–0.34 points/year**; steepest in the 60s.
  - Qmax falls ≈ **2.1%/year**; rapid decline predicted only by age ≥70 and Qmax <10 mL/s.
  - Volume grows ≈ **0.6 mL/year** (≈1.9%/year), faster with higher baseline PSA and larger volume.
- Correlations are somewhat stronger in the community: volume >50 mL ≈ 3.5-fold odds of moderate–severe LUTS.
- Serum PSA and volume have a strong log-linear relationship (older men have more volume per unit PSA) but cannot predict volume accurately in an individual.
- Symptom change needed to **perceive** improvement rises with baseline severity (slight improvement ≈ −2 points at low baseline vs ≈ −6 at high baseline).

6.3 Risk factors for progression

Predictor	Predicts
Age	Everything; AUR incidence rises almost linearly 40 → 80+
PSA ≥1.4–1.5 ng/mL	Volume growth, symptom/flow deterioration, AUR, surgery
Volume >30–40 mL	AUR (≈3-fold), surgery, incident LUTS
IPSS ≥8 / high bother	AUR, surgery (bother drives elective crossover)
Qmax <10–12 mL/s	AUR, faster flow decline, treatment
Raised post-void residual, incomplete emptying	Treatment failure, AUR

- Placebo arm of **Medical Therapy of Prostatic Symptoms (MTOPS)**: overall progression **4.5 per 100 patient-years** (17% at 4 years); mostly **IPSS rise ≥4** (14%); AUR 2%; invasive therapy 5%; **no renal insufficiency**.

- Placebo arm of **Proscar Long-Term Efficacy and Safety Study (PLESS)**: symptoms and Qmax drift back to baseline after the initial placebo response, fastest in the highest PSA tertile (3.3–10 ng/mL); surgery 10% over 4 years (2.5%/year), rising from ≈6% to ≈15% across PSA or volume tertiles.
- Placebo arm of **Reduction by Dutasteride of Prostate Cancer Events (REDUCE)** (no LUTS placebo bias): volume 40–80 mL → hazard ratio 1.67 for incident LUTS; Qmax <10 mL/s → 1.67.
- Web-based nomograms (dutasteride trial data) predict IPSS change, AUR and surgery from age, IPSS, Qmax, residual, volume and PSA.

> **KEY POINT** Why use PSA? It is a cheap surrogate for volume, and volume is the main driver of AUR and surgery. Among >100 candidate predictors, a multi-variable model was only slightly better than PSA alone.

7 · COMPLICATIONS

> **KEY POINT** Serious complications (death, renal failure, stones, infection) have become rare. What remains common is **AUR and surgery**, both predicted by the same factors: age, symptoms, low flow and a large gland (PSA).

7.1 Acute urinary retention

- One of the most significant outcomes; historically 25–33% of TURP indications; most men failing catheter removal still undergo surgery.
- Mechanisms: obstructive, myogenic and neurogenic; precipitants include infection, overdistension, excess fluid, alcohol, sexual activity, debility and bed rest. Prostatic infarction is **not** a proven cause.

	Spontaneous AUR	**Precipitated AUR**
Trigger	None	Non-prostatic surgery, anaesthesia, catheterisation, sympathomimetics, anticholinergics, antihistamines
Later spontaneous AUR	15%	9%
Later surgery	**75%**	**26%**

- Incidence rises with selection: community ≈ **4.5–6.8/1000 person-years** (Physicians' Health Study, Olmsted County); watchful waiting arm of the Veterans Affairs (VA) trial 9.6; PLESS placebo 18; MTOPS placebo 6. Hazard is constant over time.
- **Age × symptoms multiply**: Olmsted men in their 70s with IPSS >7 → 34.7/1000 person-years.

Olmsted County risk factor	**Relative risk of AUR**
IPSS ≥8	3.2
Qmax <12 mL/s	3.9
Volume >30 mL	3.0
Age ≥70 (with symptoms/low flow)	up to ≈15

- PSA >1.4 ng/mL → several-fold AUR risk; volume >40 mL ≈ 3-fold (pooled trials); risk rises almost linearly with PSA for spontaneous and precipitated AUR.
- Best IPSS item predictors: incomplete emptying, voiding again within 2 hours, weak stream. Recurrence risk: older age, diabetes, neurological disease.

7.2 Renal impairment / upper tract deterioration

- Older surgical series: ≈1 in 7 presenting for TURP had renal failure; renal failure raises TURP complications and mortality (up to 6-fold).

- "**Silent prostatism**": asymptomatic men presenting with renal failure from BOO; rare but important.
- Modern cohorts: **no** renal insufficiency due to BPH in >3000 MTOPS men over 4 years (screening may have excluded high-risk men).
- † Management: catheter drainage first, watch for post-obstructive diuresis, allow creatinine to stabilise, then definitive surgery.

7.3 Bladder stones

- Autopsy: ≈ **8×** commoner with BPH, with **no** increase in renal/ureteric stones (stasis, not metabolic). Clinical risk small; screen if haematuria or stuttering stream.

7.4 Haematuria

- Gross haematuria/clots from BPH alone; associated with ↑ prostatic microvessel density. **Finasteride** is reasonable first-line (modulates VEGF). † Exclude urothelial cancer and stones first (cystoscopy, upper-tract imaging).

7.5 Urinary tract infection (UTI) and incontinence

- UTI: main surgical indication in ≈1/8 of older series; residual–UTI link intuitive but unproven; MTOPS placebo 0.1/100 patient-years.
- Incontinence: **overflow** (overdistension) or **urge** (DO in up to half of obstructed men); low rates in trials.

7.6 Mortality and the window for cure

- BPH mortality has collapsed since the 1950s. Nocturia and frequency (storage LUTS) are associated with ↑ all-cause mortality.
- No direct evidence that delay causes irreversible damage; indirect evidence = **lesser benefit of delayed TURP** in the VA trial.

7.7 Surgery as an end point

- AUR = hard end point (like a fracture); surgery = soft end point, varying with provider, region and era (like coronary bypass after infarction). Most surgery is for symptoms.
- Olmsted County: moderate–severe symptoms, Qmax <12, volume >30 mL, PSA ≥1.4 each ≈ 4× treatment risk; ≈1 man in 4 treated in his eighth decade.
- Baltimore Longitudinal Study of Aging: cumulative surgery by number of risk factors (age, incomplete emptying, change in stream, DRE enlargement): 0 → 3%, 3 → 37%.

8 · LANDMARK TRIALS

Trial	Design	Result	Take-home
VA Cooperative Study	TURP vs watchful waiting, moderate symptoms	Failure halved (relative risk 0.48); 5-yr failure 10% vs 21%; ≈36% crossed over (mostly elective, high bother); crossover TURP gave smaller gains	Watchful waiting is safe in moderate LUTS; delayed surgery may give less benefit
PLESS	Finasteride vs placebo, 4 yr, enlarged gland	Placebo: symptoms/Qmax return to baseline; AUR and surgery rise with PSA/volume tertile	PSA/volume predict progression, AUR and surgery
MTOPS	Placebo vs doxazosin vs finasteride vs combination, 4–5 yr	Placebo progression 4.5/100 patient-years, mostly IPSS ≥4 rise; no renal failure	Defined composite "clinical progression"; risk rises with PSA and volume
Olmsted County Study	Population cohort, men 40–79	IPSS +0.2–0.34/yr, Qmax −2.1%/yr, volume +0.6 mL/yr; AUR 6.8/1000 person-years	Best natural-history data; age, IPSS, Qmax, volume predict AUR and treatment

9 · MCQ ONE-LINERS

1. BPH is hyperplasia (↑ cell number), not hypertrophy.
2. All BPH nodules arise in the transition zone or periurethral region.
3. The middle lobe is periurethral in origin.
4. Earliest periurethral nodules are stromal, resembling embryonic mesenchyme.
5. No histological BPH below age 30; autopsy prevalence 88% in the 80s.
6. Autopsy prevalence is similar across ethnic and geographic groups.
7. Prepubertal castrates never develop BPH; androgens are permissive.
8. Intraprostatic DHT is maintained, not raised, in BPH.
9. Type 2 5α-reductase sits in stromal interstitial fibroblasts; SRD5A2 on chromosome 2p23.
0. Finasteride inhibits type 2; dutasteride types 1 and 2.
1. About 90% of prostatic androgen is testicular-derived DHT.
2. ERα is stromal and proliferative; ERβ is epithelial and antiproliferative.
3. FGF-7 (KGF): stromal source, receptor only on epithelium.
4. TGF-β inhibits epithelium but stimulates fibroblasts.
5. First prostatic growth factor identified: basic FGF (FGF-2).
6. Familial BPH: autosomal dominant, early onset, large gland, normal 5-ARI response.
17. Only humans and dogs develop BPH; the dog rarely obstructs (no capsule).
18. TUIP relieves obstruction without reducing volume — evidence for the capsule's role.
19. Dynamic component is mediated by α1a-adrenoceptors.
20. Trabeculation reflects increased detrusor collagen.
21. About one-third keep storage symptoms after relief of obstruction.
22. IPSS bands: mild 0–7, moderate 8–19, severe 20–35; lacks incontinence and pain items.
23. DRE underestimates prostate volume, more so in large glands.
24. Olmsted: prostate grows ≈0.6 mL/year; Qmax falls ≈2.1%/year.
25. About 40% improve on watchful waiting or placebo.
26. Trial "placebo effect" partly reflects regression to the mean.
27. MTOPS placebo progression 4.5/100 patient-years, mostly symptomatic.
28. No BPH-related renal insufficiency occurred in MTOPS.
29. Spontaneous AUR leads to surgery in 75%; precipitated AUR in 26%.
30. PSA >1.4 ng/mL and volume >40 mL predict AUR and surgery.
31. Bladder stones are 8× commoner with BPH; renal stones are not.
32. Finasteride is first-line for BPH-related haematuria.
33. Vasectomy is not a risk factor for BPH.

150 Evaluation and Nonsurgical Management of Benign Prostatic Hyperplasia

Reading time ≈ 73 min (slow pace)

1 · DEFINITIONS AND THE DIAGNOSTIC PROBLEM

- **Benign prostatic hyperplasia (BPH)** is now a **histological** diagnosis: proliferation of smooth muscle and epithelial cells in the **transition zone**.
- **Lower urinary tract symptoms (LUTS)**: umbrella term for dysfunction of the storage and voiding phases of micturition.
- Causal chain: BPH (histology) → **benign prostatic enlargement (BPE) → benign prostatic obstruction (BPO) → bladder outlet obstruction (BOO)** → LUTS.
- LUTS are multifactorial: detrusor overactivity (DO), detrusor underactivity (DUA), urethral stricture, urinary tract infection (UTI), and hormonal, metabolic, cardiovascular and neurological disease.

> **KEY POINT** A man with bothersome LUTS cannot be assumed to have BPH-related BOO until fully assessed. The work-up has two aims: **grade severity and bother**, and **identify the cause**, because the cause dictates treatment. Every test is judged by one question: can it separate outlet obstruction from bladder dysfunction? Symptoms, flow, postvoid residual (PVR), prostate volume and cystoscopy all fail; only the **pressure–flow study** passes, so it is the gold standard, reserved for men in whom the answer changes surgery.

Governing guidelines: American Urological Association (AUA) BPH guideline (2021); European Association of Urology (EAU) non-neurogenic male LUTS guideline (2022).

2 · HISTORY AND SYMPTOM ASSESSMENT

Medical history

- **Comorbidities**: diabetes mellitus and insipidus; renal disease, heart failure, sleep apnoea; neurological disease (Parkinson disease, multiple sclerosis, cerebrovascular disease, spinal cord injury, disc prolapse).
- Ask about haematuria and prior lower-tract surgery.
- **Drugs**: loop diuretics and β-blockers worsen storage symptoms; anticholinergics impair bladder contractility.
- **Lifestyle**: excess fluid and caffeine worsen storage symptoms.
- Assess **sexual function** — LUTS/BPH are strongly linked to sexual dysfunction.
- Reassure: LUTS do **not** raise prostate cancer (PCa) risk. The apparent association is detection bias (men treated for LUTS are biopsied more); in the Prostate Cancer Prevention Trial (PCPT) placebo arm LUTS were not associated with PCa.

Voiding (obstructive)	Storage (irritative)
Weak stream, hesitancy, straining, intermittency, incomplete emptying, prolonged micturition, dribbling	Frequency, urgency, urge incontinence, nocturia — now grouped as **overactive bladder (OAB) syndrome**
Often due to BOO, but not always BPE/BPH	May also result from BPE/BPH obstruction

Symptom scores

- **AUA Symptom Index (AUASI)**: 7 questions — frequency, nocturia, weak stream, hesitancy, intermittency, incomplete emptying, urgency — each 0–5, total **0–35**.
- **International Prostate Symptom Score (IPSS)**: World Health Organization (WHO) adaptation (1993) = the 7 AUASI questions + **one bother/quality-of-life (QoL) question** (†scored 0–6).

International standard; storage (3 questions) and voiding (4 questions) subscores are not yet validated.

Band	IPSS / AUASI
Mild	**0–7**
Moderate	**8–19**
Severe	**20–35**

- Neither score **diagnoses BOO**: symptom severity correlates poorly with urodynamic obstruction.
- Minimum change the patient perceives: **3 points**. Men with severe baseline symptoms need proportionately smaller change, so **percentage** reduction is more reliable.
- Other instruments: **BPH Impact Index (BII)** (4 questions on impact); **Visual Prostate Symptom Score (VPSS)** — pictures, for elderly/low-literacy men; Overactive Bladder Symptom Score (OABSS)/OAB-q — detect occult storage LUTS.

3 · PHYSICAL EXAMINATION AND DIGITAL RECTAL EXAMINATION

- **Suprapubic**: palpable or percussible bladder (retention).
- **Penis**: meatal stenosis (itself causes BOO), phimosis, penile lesions, ventral urethral mass (stricture), discharge (urethritis).
- **Scrotum**: orchiepididymitis; small testes and reduced secondary sex characteristics suggest low testosterone, which accompanies central obesity and metabolic syndrome (associated with BPE — lifestyle and diet help).
- **Neurological**: perineal and lower-limb motor/sensory function; **anal tone** — flaccid or spastic suggests neurological disease.
- **Digital rectal examination (DRE)** — two aims: baseline prostate volume (PV) and exclusion of PCa (nodule, induration). Low sensitivity for PCa; combined with PSA it reduced overdiagnosis in ERSPC.
  - DRE **underestimates PV above 30 mL** (lateral lobes partly impalpable), but its positive predictive value is good at 30 mL and near-perfect at 50 mL; a flat/normal gland predicts PV <30 mL.
  - PV does not correlate with symptom severity or degree of obstruction, but the **30 mL** and **50 mL** cut-offs predict treatment response and progression.

4 · FREQUENCY–VOLUME CHARTS AND NOCTURIA

International Continence Society (ICS) diary type	Records
Micturition chart	Time of each void only, ≥24 h
Frequency–volume chart (FVC)	Time and volume of each void, day and night, 24 h
Bladder diary	FVC + fluid intake, incontinence episodes, pads, urgency

- **Indication**: basic evaluation of predominant **storage LUTS or nocturia**. FVCs are the mainstay of nocturia assessment and more accurate than recall (IPSS).
- **Duration**: reference **3 days** — adherence falls with 7 days, reliability rises with more days. Keeping a diary has a bladder-training effect; tell the patient not to change his routine.
- Derived measures: 24-h frequency, mean voided volume, **nocturnal polyuria index** (fraction of 24-h urine produced at night).
- **Nocturia** (ICS): waking ≥1 time at night to void; a storage symptom.
- **Nocturnal polyuria** (ICS): nocturnal output **>20%** of 24-h output in younger men, **>33%** over 65 years.

ALGORITHM

Nocturia work-up: FVC → classify: †global polyuria (>40 mL/kg/24 h → look for diabetes mellitus/insipidus, excess intake) / nocturnal polyuria (heart failure, sleep apnoea, evening fluids, diuretic timing) / reduced bladder capacity (OAB, BOO) → treat cause → if persistent, look for other causes such as **sleep apnoea** (AUA).

Treatment of nocturnal polyuria: education and lifestyle (no fluid 2 h before bed, avoid evening alcohol/caffeine, retime diuretics) ± **desmopressin** (vasopressin analogue), per nocturia guidelines. †Main risk is **hyponatraemia** (elderly): check sodium at baseline, within the first week and at 1 month.

5 · LABORATORY TESTS

Urinalysis — mandatory in every man with LUTS

- Dipstick and/or microscopy: detects UTI, microhaematuria, glycosuria, proteinuria.
- Pyuria + nitrite → UTI (but insensitive in painless LUTS; reference 10^5 colony-forming units (CFU)/mL).
- Gross haematuria → exclude bladder cancer. Glycosuria/proteinuria/ketonuria → diabetes.
- **Urine cytology**: severe storage symptoms with dysuria, especially smokers (carcinoma in situ).

Prostate-specific antigen (PSA)

- Three roles: **PCa risk**, **PV surrogate**, **predictor of BPH progression**.
- **Indication**: only if a PCa diagnosis would change management — i.e. life expectancy **>10 years**. Counsel on false positives/negatives and biopsy complications.
- **Volume surrogate**: area under curve ≈0.76–0.78. Age-specific PSA cut-offs predicting **PV >40 mL**: **>1.6** (50s), **>2.0** (60s), **>2.3 ng/mL** (70s). Predicts transrectal ultrasound (TRUS) volume within ±20% in most men.
- **Outcome predictor**: baseline PSA consistently predicts **acute urinary retention (AUR)** and BPH surgery in placebo arms; also predicts change in symptoms and maximum flow rate (Qmax).
- **5α-reductase inhibitors (5-ARIs) reduce PSA by 40–50% by 12 months** → always obtain a pretreatment baseline. †Rule of thumb: double the measured PSA after ≥6 months of 5-ARI; any rise from the on-treatment nadir needs evaluation.

Renal function

- **EAU**: creatinine/estimated glomerular filtration rate (eGFR) when hydronephrosis, renal insufficiency or retention is suspected. **AUA**: not explicitly recommended.
- Renal failure is rare (<1% over 4 years in MTOPS); Qmax correlates with eGFR, PV/PVR/IPSS do not. Renal insufficiency raises post-surgical complications.

6 · INSTRUMENTAL INVESTIGATIONS

Postvoid residual (PVR)

- Urine left after micturition; no standard normal. Measure soon after voiding by **ultrasound**; catheterisation is the gold standard but invasive.
- Thresholds: **<30 mL** insignificant; persistently **>50 mL** important; **200–300 mL** may indicate DUA and predicts poorer response; **>300 mL** = significant chronic retention.
- Variable; a high PVR may reflect BOO **or DUA**.
- Predicts symptomatic deterioration (MTOPS >39 mL; ALTESS >93 mL) but **not surgery**. Rising PVR over time predicted AUR (MTOPS placebo); PVR ≥500 mL predicted AUR recurrence.
- Measure at baseline and follow-up; men with significant PVR on non-surgical therapy need close monitoring.

Uroflowmetry

- Non-invasive; integrates bladder and outlet. Key parameters: **Qmax** (peak flow), voided volume, flow pattern. Average flow is less accurate than Qmax.
- Errors: inaccurate when voided volume **<150 mL** (125–150) → repeat; single flow underestimates achievable Qmax; void in preferred position, interpret only voids the patient considers representative.
- **Core limitation**: cannot distinguish BOO from DUA (DUA in 9–48% of men having urodynamics for non-neurogenic LUTS).
- Cut-offs: Qmax **10 mL/s** — higher specificity; **15 mL/s** — higher sensitivity. Qmax <15 mL/s is the working definition of obstruction but does not exclude decompensation.
- Optional at initial assessment; **recommended before any active treatment** and for monitoring.

Urodynamics

- **Filling cystometry** (storage): detects DO — common in BPH, increases with age and obstruction; identifies men likely to benefit from antimuscarinics.
- **Pressure–flow study (PFS)** (voiding): detrusor pressure (y-axis) vs flow (x-axis). BOO = **low flow + high pressure**; DUA = **low flow + low pressure**.
- DUA predicts long-term symptomatic failure and lower flow after surgery. Bladder contractility index neither improves after surgery nor declines with conservatively managed BOO — contractility is independent of treatment and duration of obstruction.
- **Not routine** (EAU, AUA): a randomised trial of adding urodynamics did not change IPSS or surgery rates at 18 months.

Indications for PFS before invasive treatment (EAU)
Previous unsuccessful invasive treatment
Cannot void **>150 mL**
PVR **>300 mL**
Age **>80** or **<50 years** with predominantly voiding LUTS
Suspected neurological disease, previous pelvic surgery, suspected DO/DUA
AUA: may be done when **Qmax >10 mL/s** before surgery (an unobstructed bladder may be the cause)

Imaging

- **Upper tracts**: routine ultrasound not recommended (hydronephrosis rare). Indicated for raised creatinine, large PVR, haematuria, UTI, stones, prior tract surgery. Intravenous urography is obsolete.
- **Prostate**: suprapubic ultrasound or **TRUS (method of choice for PV)**. Required before a 5-ARI (efficacy depends on volume) and before **any BPH surgery** to choose the technique. A protruding **median lobe** influences the choice of minimally invasive procedure.
- PV predicts progression and AUR (MTOPS; ALTESS placebo PV >49 mL; Olmsted County) but correlates weakly with BOO.

Ultrasound surrogate of BOO	**Method / cut-off**	**Status**
Intravesical prostatic protrusion (IPP)	Tip of protrusion to base, sagittal suprapubic view, bladder 100–200 mL; **≥10 mm** = high grade	Reasonable accuracy; predicts need for medical therapy and **failed trial without catheter (TWOC)**. Not a substitute for PFS
Bladder wall thickness (BWT)	Mucosa to adventitia; 2 mm	Predicts successful TWOC; not recommended (hypertrophy multifactorial)
Detrusor wall thickness (DWT)	Hypoechoic detrusor only	Among the most accurate; not recommended

Ultrasound surrogate of BOO	Method / cut-off	Status
Ultrasound-estimated bladder weight (UEBW)	Corrects BWT for filling	Two discordant studies

Other non-invasive tests: **penile cuff test** (cuff pressure that interrupts flow ≈ isovolumetric bladder pressure; high sensitivity), Doppler detrusor resistive index, near-infrared spectroscopy (NIRS). Highest accuracy: penile cuff, DWT, UEBW, NIRS — but **none is recommended as an alternative to PFS**.

Cystourethrography and cystoscopy

- Voiding cystourethrogram: suspected stricture, diverticulum, reflux.
- **Cystourethroscopy** shows prostatic/bladder-neck anatomy, trabeculation (grade 0–3), diverticula but **neither diagnoses BOO nor decides treatment** (obstructed men may have a normal wall).
- Indications: gross haematuria or history of bladder cancer; recurrent UTI or urethral injury (stricture); previous prostate/urethral surgery; surgical planning — **median lobe must be excluded before prostatic urethral lift (PUL)**; young man with small prostate → exclude **bladder-neck contracture**.

7 · EVALUATION ALGORITHMS

ALGORITHM

EAU (men ≥40 years): history (incl. sexual function) → symptom score → examination + DRE → urinalysis → PVR → PSA (if life expectancy >10 years and result changes management).
Abnormal DRE, high PSA, abnormal urinalysis, suspected neurological disease → manage under relevant guideline.
Not bothersome + normal tests → no further work-up. Significant PVR → renal ultrasound ± renal function.
Bothersome → FVC (storage/nocturia) → prostate ultrasound + uroflow if treatment contemplated → medical algorithm. Surgery contemplated → endoscopy if it alters choice; PFS for specific indications.

ALGORITHM

AUA: Step 1 history, examination, IPSS, urinalysis ± PVR/flow; **PVR >300 mL → manage as non-neurogenic chronic urinary retention** regardless of symptoms → Step 2 counsel: behavioural/lifestyle, medical therapy, procedural referral → Step 3 reassess at **4–12 weeks**: improved → routine follow-up; nocturia predominant → FVC + drug trial; incomplete response → FVC, PVR/flow, urodynamics, cystoscopy → OAB pathway, surgical algorithm (BOO), or both (mixed).

8 · CONSERVATIVE MANAGEMENT — WATCHFUL WAITING (WW)

- **Indication**: mild-to-moderate, uncomplicated LUTS with minimal bother.
- **Excluded** by complicating factors: gross haematuria, UTI, raised PSA, raised PVR, neurological disease.
- Components: **reassurance, education, lifestyle intervention, periodic monitoring**.
- **Rationale**: natural history is mostly benign — AUR and surgery are uncommon and about half of men are stable over 15 years. Progression = symptom or flow deterioration, AUR, or need for surgery.
- Veterans Affairs trial (transurethral resection of the prostate [TURP] vs WW): TURP better for symptoms, flow and failure, but most WW men did not fail; about a quarter crossed over, predicted by high baseline **bother**; AUR and renal failure rare at 5 years.

- **Self-management** (three small-group education/lifestyle sessions) reduced treatment failure and symptom scores versus standard care in randomised trials.
- Review **yearly**; progression or complications → active treatment.

Domain	Advice
Fluids	**1.5–2 L/day** adjusted for climate/activity (guided by FVC); restrict before journeys; none for 2 h before bed
Caffeine, alcohol	Caffeine-free drinks; avoid evening alcohol; small-volume drinks instead of beer
Medication	Retime diuretics; substitute where possible
Toileting	**Double voiding**; **urethral milking** for postmicturition dribble; avoid constipation
Bladder retraining	Distraction (mental tasks, perineal pressure, pelvic floor contraction); suppress urge 1 → 5 → 10 min, increasing weekly; target ≥3 h between voids or 200–400 mL voided volume; monitor with FVC

> **KEY POINT** Decision to treat is driven by **bother and complicating features**, not by symptom score or gland size alone.

9 · MEDICAL THERAPY — INDICATIONS AND CHOICE

- First line for bothersome LUTS **without an absolute indication for surgery**.
- **Absolute indications for surgery**: refractory/recurrent AUR, recurrent UTI, renal insufficiency, bladder stones, recurrent gross haematuria → *Ch 151*.
- Tailor to profile: voiding → α1-blocker, 5-ARI or phosphodiesterase type 5 inhibitor (PDE5I); storage → antimuscarinic or β3-agonist; nocturnal polyuria → lifestyle ± desmopressin.

ALGORITHM

EAU medical algorithm: not bothersome → WW ± lifestyle | nocturnal polyuria → lifestyle ± vasopressin analogue | storage predominant → lifestyle ± antimuscarinic or β3-agonist | **PV >40 mL and long-term treatment intended** → 5-ARI ± α1-blocker or PDE5I | otherwise → α1-blocker or PDE5I → residual storage symptoms → add antimuscarinic or β3-agonist.

AUA: α-blocker first (or PDE5I first if erectile dysfunction [ED] coexists) → **prostate >30 mL** → consider adding 5-ARI → inadequate response → trial of PDE5I → still inadequate → discuss surgery.

> **KEY POINT** **Reading drug trials**: placebo improves both scores and flow (ideal: 4-week placebo run-in); efficacy is judged by the 3-point/percentage threshold; Qmax gains should be expressed as percentages. Treatment goals: relieve symptoms, reduce BOO, improve emptying, and **prevent progression**.

10 · A1-ADRENERGIC BLOCKERS

Rationale

- Noradrenaline acting on α1-receptors (mainly **α1a, ~70%** of prostatic α1) in prostate, bladder neck and prostatic urethra creates the **dynamic component** of BOO. Blockade relaxes smooth muscle → rapid relief. They do **not** shrink the gland.
- Bladder effects: α1d on detrusor, urothelium and spinal afferents — blockade may increase capacity, reduce DO, and improve bladder perfusion (ischaemia contributes to OAB).

Subtype	Main sites	Clinical relevance
α1a	Prostate, bladder neck, prostatic urethra; **iris dilator**	Dynamic obstruction; **ejaculatory dysfunction**; **intraoperative floppy iris syndrome (IFIS)**
α1b	Blood vessels	**Vascular AEs** — hypotension, dizziness

Subtype	Main sites	Clinical relevance
α1d	Detrusor (twice α1a), urothelium, spinal cord	Storage symptoms

Agents (Table 150.1)

Class	Drug	Dose	Notes
Non-selective	Phenoxybenzamine	10 mg bd	α1+α2; serious cardiovascular AEs — abandoned
Selective short-acting	Prazosin, indoramin	bd	No longer used for LUTS
Selective long-acting (quinazolines)	Terazosin	5–10 mg od	Start **1 mg at bedtime**, titrate
	Doxazosin IR	2–8 mg od	Start 2 mg at bedtime, titrate
	Doxazosin GITS (sustained release)	4–8 mg od	Start **4 mg** after breakfast, **no titration**
Uroselective	Alfuzosin ER	10 mg od	Clinically (not receptor) uroselective; same meal daily
	Tamsulosin	0.4 mg od (to 0.8)	α1a/α1d; after same meal daily
	Silodosin	8 mg od	Most α1a-selective; with a meal
	Naftopidil	25–75 mg/day	Only α1d-selective; Asia only

> KEY POINT **Why titrate terazosin/doxazosin?** α1b blockade causes **first-dose hypotension** and syncope; bedtime starting dose and titration avoid it. Uroselective agents are fixed-dose. Trap: doxazosin GITS starts at 4 mg without titration.

Efficacy

- Level 1a evidence for every agent; all lower the BOO index (silodosin most — subgroup data). Roughly double the placebo symptom reduction with modest Qmax gain; benefit **durable** (PREDICT 1 year, MTOPS 4 years).
- **Onset fast**: tamsulosin raises flow within hours and improves symptoms by 1 week; silodosin symptoms in 3–4 days.
- Dose ceilings: tamsulosin **0.4 mg** is the efficacy plateau (0.8 mg adds only AEs; 0.2 mg not significant); alfuzosin 15 mg no better than 10 mg; terazosin best at 10 mg (but most dizziness).
- Tamsulosin relieves storage symptoms too (nocturia). Naftopidil gives more storage relief (α1d) than low-dose tamsulosin.
- **Head-to-head: no α1-blocker is clearly more effective** (terazosin = doxazosin; alfuzosin = tamsulosin; silodosin = tamsulosin 0.4 mg). They differ in side effects.
- Doxazosin alone reduced overall progression 39% in MTOPS, but **not AUR or surgery**.

Adverse effects

- **Commonest**: dizziness, asthenia, (orthostatic) hypotension, anejaculation.
- **Vascular**: more with terazosin/doxazosin (α1b). Symptomatic orthostatic hypotension = standing systolic fall ≥20 mm Hg with symptoms. BP fall clinically significant only in untreated/poorly controlled hypertensives. Alfuzosin slightly more than tamsulosin, especially >75 years and with antihypertensives; silodosin no excess.
- **ALLHAT**: doxazosin arm stopped (excess heart failure vs chlorthalidone) — do not use α-blockers as sole antihypertensive. Use cautiously with antihypertensives; **never combine two α-blockers**.
- **Elderly**: dizziness → falls, fractures; new tamsulosin/alfuzosin/silodosin use increased falls and fractures; severe hypotension needing admission is most frequent in the **first 8 weeks**. Warn every patient; check BP in the first week.

- **Sexual**: ED incidence equals placebo; no effect on libido. **Ejaculatory dysfunction** mechanisms: bladder-neck relaxation (retrograde), seminal vesicle/vas effect (**anejaculation — emission failure**), central (D2/5-HT1A, tamsulosin). Ranking: silodosin highest (OR ≈32; real-life most men) > tamsulosin (dose-dependent; OR ≈9) > alfuzosin (lowest of uroselectives); doxazosin/terazosin minimal. Younger men are more affected.

> **KEY POINT** **IFIS** — triad: **flaccid billowing iris, iris prolapse toward incisions, progressive miosis** despite dilation during cataract surgery. Cause: α1a blockade of iris dilator → dilator atrophy, so it occurs **even years after stopping**; stopping the drug pre-operatively does **not** reduce risk. Highest with **tamsulosin**, then alfuzosin, doxazosin, naftopidil. Tell the patient to inform the ophthalmologist. †In a man awaiting cataract surgery, defer starting an α1-blocker until after the operation.

Drug	Vascular	Ejaculatory	IFIS	Titration
Terazosin	High	Low	Reported	Yes
Doxazosin	High (GITS lower)	Low	Lower	IR yes; GITS no
Alfuzosin	Intermediate	Lowest of uroselectives	Intermediate	No
Tamsulosin	Low	Dose-dependent	**Highest**	No
Silodosin	Lowest	**Highest**	Not well studied	No

> **KEY POINT** One efficacy, three toxicity profiles: **α1b → vascular** (terazosin, doxazosin); **α1a → ejaculation and IFIS** (silodosin, tamsulosin); **α1d → storage benefit** (naftopidil). Choosing an α-blocker is choosing the side effect the patient tolerates best.

11 · 5α-REDUCTASE INHIBITORS (ANDROGEN MANIPULATION)

Rationale

- Prostate is androgen dependent; **dihydrotestosterone (DHT)** is more potent than testosterone (dissociates more slowly from the androgen receptor).
- Isoenzymes: **type 2** — prostate/genital tissue; **type 1** — liver, skin.
- Lowering DHT shrinks the gland → relieves the **static component**. 5-ARIs act without disturbing the hypothalamic–pituitary–gonadal axis — hence they are the only androgen-manipulating class in use. Castration, cyproterone, flutamide, bicalutamide, gonadotropin-releasing hormone analogues (cetrorelix), zanoterone, aromatase inhibitors: no useful efficacy or unacceptable toxicity (gynaecomastia, breast pain, diarrhoea, loss of libido).

	Finasteride	Dutasteride
Isoenzyme	Type 2	Types 1 and 2
Serum DHT	Partial fall (†~70%)	~90% fall
Dose	**5 mg** daily	**0.5 mg** daily
PV reduction	~18–20%	~26%
AUR / surgery reduction	57% / 55% (PLESS)	57% / 48% (phase III)
Prevention trial	PCPT	REDUCE
Head to head	No difference in symptoms, flow, PV or AEs	

Key clinical facts

- **Slow onset**: maximum PV reduction at **6 months**; perceptible benefit needs ≥6 months (trials of 6–12 months can miss benefit).
- **Volume dependent**: best when prostate **>40 mL** (PLESS PV >55 mL → 70% risk reduction). Guideline indication: enlarged prostate (EAU >40 mL; AUA >30 mL) with moderate-to-severe LUTS.

- Modest symptom/flow gain, but the **only drug class that alters natural history** — reduces AUR and surgery. Placebo glands keep growing.
- **PSA halves** (from 3 months; ~50% at 6–12 months) — baseline essential. 5-ARIs actually **improve PSA's accuracy** for predicting PCa.
- PCPT (finasteride) and REDUCE (dutasteride; 23% relative reduction): fewer cancers but a small increase in high-grade (Gleason ≥8) disease → **not licensed for PCa prevention**. The high-grade excess may be a volume artefact (smaller gland → better biopsy sampling).
- Finasteride reduces recurrence of **refractory BPH-related haematuria**.

Adverse effects

- **Sexual**: ED, decreased libido, reduced ejaculate volume — about double placebo; equal for both drugs; mostly wane after the first 1–2 years. Mechanism: low DHT reduces cavernosal nitric oxide synthase and causes structural change.
- **Depression**: real signal — FDA label (2011); more depression and self-harm (not suicide) in cohorts. Counsel.
- Gynaecomastia (rare). Diabetes, cardiovascular disease, bone loss: **not confirmed**.
- **Post-finasteride syndrome**: persistent sexual, physical and neurological symptoms after stopping; poorly understood (some consider it nocebo).

> **KEY POINT** 5-ARIs treat the gland, not the symptom: slow, volume-dependent, modest symptomatically, but they prevent AUR and surgery — and they halve PSA, which is both a trap and a tool.

12 · ANTIMUSCARINICS FOR STORAGE LUTS

- Parasympathetic acetylcholine acts on detrusor **M2/M3** receptors; blockade raises the contraction threshold and capacity, acting mainly on **afferent (sensory)** pathways.
- **Indication**: moderate-to-severe LUTS with predominant storage symptoms, **baseline PVR <200 mL**, no history of AUR. Use cautiously because of the theoretical risk of retention.
- Agents (once-daily ER preferred): darifenacin 7.5/15 mg, fesoterodine 4/8 mg, oxybutynin (IR 5 mg 3–4×/day; ER), propiverine, solifenacin 5/10 mg, tolterodine (ER 4 mg), trospium.
- **Monotherapy** in BPH: controversial — no fall in total IPSS or change in Qmax; storage relief proven. Favoured by small prostate, low PSA (<1.3 ng/mL), higher Qmax and high storage subscore.
- **Safety in BOO**: small PVR rise, AUR no commoner than placebo — but every trial excluded PVR ≥200 mL.
- AEs: **dry mouth** (commonest), constipation, dizziness, cognitive impairment/delirium. **Oxybutynin** crosses the blood–brain barrier most → avoid in elderly/cognitively impaired; trospium (quaternary amine) least†.

> **KEY POINT** Antimuscarinics treat the bladder, not the outlet, so every recommendation is anchored to a PVR ceiling (200 mL) — the question is how well the detrusor empties.

13 · B3-AGONISTS

- β3 is the predominant bladder β-receptor. Stimulation → adenylate cyclase → ↑cyclic AMP → protein kinase A → ↓calcium → **detrusor relaxation during storage without weakening the voiding contraction** → no change in Qmax or detrusor pressure at Qmax.
- **Mirabegron**: start **25 mg** → **50 mg** daily. Urodynamic safety trial in BOO: Qmax unchanged; PVR rose only at 100 mg. Lacks anticholinergic burden (dry mouth less than solifenacin/tolterodine).
- AEs: nasopharyngitis, headache, hypertension (similar to placebo; BP rise ≤1 mm Hg). **Contraindicated in severe uncontrolled hypertension** (**≥180/110 mm Hg**); measure BP before and during treatment; QT prolongation at 200 mg. Moderate CYP2D6 inhibitor †.

- **Vibegron** 75 mg daily: highly selective, half-life 25–38 h, no CYP2D6 inhibition (fewer interactions); evidence mainly in women (EMPOWUR).
- Guideline (EAU, AUA): option for moderate-to-severe LUTS with predominant storage symptoms.

14 · PHOSPHODIESTERASE TYPE 5 INHIBITORS

- Mechanisms (nitric oxide–cyclic GMP axis): ↑bladder/prostate perfusion and oxygenation; prostatic and bladder-neck smooth-muscle relaxation; antiproliferative (blocks fibroblast → myofibroblast change); **afferent inhibition**; anti-inflammatory (metabolic syndrome).
- **Tadalafil 5 mg once daily** is the only PDE5I licensed for LUTS (long half-life). 2.5 mg ineffective; 5 mg best benefit–risk. IPSS falls ~6 points (placebo ~3.6); benefit **independent of baseline ED**; reduces nocturia.
- **Paradox: symptoms improve but Qmax does not** — the effect is not primarily de-obstructive.
- First choice for LUTS + ED (AUA). AEs: headache, back pain, dyspepsia, flushing, myalgia. †Contraindicated with nitrates (and riociguat).

15 · COMBINATION THERAPY

α1-blocker + 5-ARI

- Short trials (6–12 months) showed no added benefit — too short for the 5-ARI, and glands were small (<40 mL).
- **MTOPS** (doxazosin, finasteride, combination, placebo; ≥4 years; no PV criterion). Progression = first of: AUASI rise **≥4**, AUR, recurrent UTI/urosepsis, incontinence, renal insufficiency (creatinine rise ≥1.5 mg/dL). Placebo 17% progressed. Risk reduction: doxazosin 39%, finasteride 34%, **combination 66%**. AUR and invasive therapy reduced only by finasteride and the combination. number needed to treat (NNT): combination 8.4; ~5 if PSA >4 and PV >40 mL.
- **CombAT** (dutasteride + tamsulosin; 4 years; high-risk entry PV ≥30 mL, PSA ≥1.5): combination reduced AUR/surgery by about two-thirds vs tamsulosin but not significantly vs dutasteride; symptoms best with combination; dutasteride beat tamsulosin long term.
- **Indication**: moderate-to-severe LUTS at high risk of progression — PV **>30–40 mL**, PSA **>1.5 ng/mL**; monotherapy doses once daily.
- **SMART**: the α-blocker can be withdrawn after 6 months of successful combination.

MTOPS risk reduction vs placebo	Doxazosin	Finasteride	Combination
Overall progression	39%	34%	66%
≥4-point symptom rise	45%	30%	64%
AUR	NS	68%	81%
Invasive therapy	NS	64%	67%

> **KEY POINT** Most progression events are symptomatic. **α-blockers prevent symptomatic progression only; 5-ARIs prevent the hard events (AUR, surgery).** The α-blocker gives rapid dynamic relief; the 5-ARI gives slow static shrinkage.

Other combinations

Combination	Indication	Caution
α1-blocker + antimuscarinic	Persistent storage LUTS after α-blocker (four-arm tolterodine trial: only combination beat placebo; solifenacin + tamsulosin OCAS — NEPTUNE)	Small PVR rise; studied only with low PVR (<150–200 mL); AUR <1–3%
α1-blocker + β3-agonist	Persistent storage LUTS (MATCH)	PVR rise; limited data; no dose adjustment

Combination	Indication	Caution
α1-blocker + PDE5I	LUTS ± ED; small extra IPSS gain	Two vasodilators → hypotension; regulators do not recommend; if needed start the second drug at lowest dose
PDE5I + 5-ARI	Enlarged prostate ± ED (tadalafil + finasteride improved IPSS and erections)	Single trial; FDA fixed-dose capsule (2021)

16 · PHYTOTHERAPY

- Plant extracts (phytosterols such as β-sitosterol, fatty acids, phytoestrogens). **Every extract is unique** — composition varies with extraction method and batch, so results are product-specific.
- Proposed mechanisms: anti-inflammatory (cyclooxygenase/lipoxygenase inhibition), in-vitro 5α-reductase inhibition (not confirmed in humans — Serenoa does not lower PSA), antiproliferative.
- **Serenoa repens** (saw palmetto): large long-term trials (ethanolic/CO_2 extracts, even at triple dose) — no benefit over placebo. **Hexane extract (Permixon) 320 mg daily with food**: fewer nocturnal voids, higher Qmax than placebo; EAU suggests it for men wishing to avoid side effects of other drugs. AEs: mild gastrointestinal; rare warfarin interaction.
- Pygeum africanum: modest short-term benefit. Hypoxis rooperi (β-sitosterol), rye pollen, nettle, pumpkin seed: unreliable evidence.
- AUA and EAU give **no general recommendation**.

17 · PROGNOSIS AND FOLLOW-UP ON MEDICAL THERAPY

- BPH is slowly progressive (Olmsted County: PV +1.9%/year).
- **Baseline risk factors for progression**: higher PSA, larger PV, greater PVR, older age, high bother. Dynamic predictors of AUR: worsening symptoms and rising PVR.
- CONDUCT: combination beat WW only in men with high baseline bother — bother predicts benefit.

ALGORITHM
Follow-up: **4–12 weeks** — response (α-blocker) and AEs → **6 months** (5-ARI response) → **annually**: symptom score, flow, PVR; PSA on 5-ARI if life expectancy >10 years. WW: yearly review.

18 · ACUTE URINARY RETENTION

- **Definition**: sudden, painful inability to void voluntarily.
- **Spontaneous** (natural history of BPH; worse TWOC prognosis; ~two-thirds) vs **precipitated** (anaesthesia/surgery, UTI, overdistension, anticholinergic or sympathomimetic drugs; better prognosis).
- Predictors of spontaneous AUR: older age, severe LUTS, low Qmax, high PVR, large PV, high PSA.
- Incidence: 2.2–8.5/1000 person-years in men; several-fold higher with LUTS.

ALGORITHM
AUR: urethral catheter decompression (†suprapubic if urethral fails) → start **α1-blocker** (e.g. alfuzosin 10 mg) → **TWOC after ~3–5 days** (median ~5) → voids: continue medical therapy (add 5-ARI if large gland), monitor → fails: re-catheterise; up to **3 TWOCs** may be tried → recurrent AUR or failure → PFS and surgery → *Ch 151*.

- Immediate surgery resolves BOO but emergency surgery has higher morbidity → **TWOC preferred** for first episodes; succeeds in about **60%**. Recurrence after successful TWOC is common, mostly within **6 months**.
- **TWOC failure predictors**: age, severe LUTS, drained volume **≥1 L**, spontaneous AUR, high-grade IPP.

- **Why α-blocker?** Increased sympathetic tone at the bladder neck (dynamic component) contributes to retention. **ALFAUR**: alfuzosin for 3 days → TWOC success 62% vs 48%. Meta-analysis: relative success ≈1.5 and less recurrence.
- **Prevention**: 5-ARIs (PLESS, dutasteride phase III, MTOPS) reduce AUR; α-blockers do not (ALTESS, MTOPS). CombAT: combination reduced AUR 68% vs tamsulosin but only 18% (NS) vs dutasteride — the 5-ARI carries the protection.

> **KEY POINT** AUR follows the two-component model: the **α-blocker relieves the dynamic component and rescues the TWOC**; the **5-ARI shrinks the static component and prevents the next episode**.

19 · LANDMARK TRIALS

Trial	Design	Result	Take-home
VA TURP vs WW	RCT, moderate LUTS	TURP better; most WW men did not fail; crossover predicted by bother	WW safe for mild–moderate LUTS
PLESS	Finasteride vs placebo, 4 years, PV ~55 mL	AUR 7→3%, surgery 10→5%; combined ↓51%	5-ARI prevents AUR/surgery
Dutasteride phase III (pooled)	2 years, PV ≥30 mL	DHT ↓90%, PV ↓26%, AUR ↓57%, surgery ↓48%	Dutasteride ≈ finasteride
MTOPS	Doxazosin/finasteride/combination/placebo, ≥4 years	Progression ↓39/34/66%; AUR and surgery only with 5-ARI	Combination for high risk; α-blocker does not prevent AUR
CombAT	Dutasteride + tamsulosin, PV ≥30, PSA ≥1.5, 4 years	AUR/surgery ↓66% vs tamsulosin; best symptoms	Combination for large gland
SMART	α-blocker withdrawal after combination	No deterioration after 6 months	Stop α-blocker at 6 months
PCPT / REDUCE	5-ARI for PCa prevention	Fewer cancers, more high grade	Not licensed for prevention
ALFAUR	Alfuzosin vs placebo before TWOC	Success 62% vs 48%	α-blocker before TWOC

20 · MCQ ONE-LINERS

1. IPSS = 7 AUASI questions + 1 QoL (bother) question; total symptom score 0–35.
2. IPSS bands: mild 0–7, moderate 8–19, severe 20–35.
3. Minimum patient-perceptible IPSS change: 3 points.
4. PSA >1.6 / 2.0 / 2.3 ng/mL (50s/60s/70s) predicts PV >40 mL.
5. 5-ARIs lower PSA by ~50% after 6–12 months; obtain baseline PSA.
6. Nocturnal polyuria: >33% of 24-h output at night in men over 65.
7. Uroflow unreliable when voided volume <150 mL.
8. Uroflowmetry cannot distinguish BOO from detrusor underactivity.
9. PFS indicated for age >80 or <50 with voiding LUTS, PVR >300 mL, voided volume <150 mL.
10. AUA: PVR >300 mL → manage as chronic urinary retention.
11. IPP ≥10 mm is high grade and predicts TWOC failure.
12. Exclude median lobe by cystoscopy before prostatic urethral lift.
13. α1a constitutes ~70% of prostatic α1-receptors.
14. Most α1a-selective agent: silodosin; only α1d-selective: naftopidil.
15. Highest IFIS risk: tamsulosin; stopping pre-operatively does not prevent IFIS.
16. Doxazosin GITS starts at 4 mg without titration.
17. Highest anejaculation: silodosin; lowest among uroselectives: alfuzosin.

18. ALLHAT doxazosin arm stopped for excess heart failure.
19. Finasteride inhibits type 2; dutasteride types 1 and 2 (DHT ↓90%).
20. PLESS: finasteride halved AUR and surgery over 4 years.
21. MTOPS combination reduced progression 66%; doxazosin did not reduce AUR.
22. CombAT entry: PV ≥30 mL and PSA ≥1.5 ng/mL.
23. Antimuscarinics in BPH: only if PVR <200 mL.
24. Mirabegron contraindicated in uncontrolled BP ≥180/110 mm Hg.
25. Tadalafil 5 mg daily improves IPSS without increasing Qmax.
26. Only phytotherapy endorsed by EAU: hexane-extracted Serenoa repens 320 mg.
27. TWOC succeeds in ~60%; α-blocker raises success (ALFAUR 62% vs 48%).
28. TWOC failure predictors: older age, drained volume >1 L, spontaneous AUR.
29. Only 5-ARIs, not α-blockers, prevent AUR.

151 Minimally Invasive and Endoscopic Management of Benign Prostatic Hyperplasia

Reading time ≈ 58 min (slow pace)

1 · SCOPE AND PRINCIPLES

- Lower urinary tract symptoms (LUTS) from benign prostatic hyperplasia (BPH): options are resection, incision, vaporisation, enucleation and minimally invasive surgical therapies (MISTs).
- **Operations in the 2023 American Urological Association (AUA) guideline**: transurethral incision of the prostate (TUIP), transurethral resection of the prostate (TURP), transurethral vaporisation of the prostate (TUVP), photoselective vaporisation of the prostate (PVP), prostatic urethral lift (PUL; UroLift), water vapour thermal therapy (WVTT; Rezūm), temporary implanted prostatic device (TIPD; iTind), robotic waterjet treatment (RWT; Aquablation), holmium or thulium laser enucleation of the prostate (HoLEP, ThuLEP), simple prostatectomy (open, laparoscopic, robotic) and prostate artery embolisation (PAE).
- Transurethral microwave therapy (TUMT) was removed in 2023 as legacy technology. The sham effect is large (sham endoscopy improves the International Prostate Symptom Score (IPSS) and maximum flow rate (Qmax)), so new devices need sham-controlled randomised controlled trials (RCTs).

> **KEY POINT** Prostate volume predicts progression (acute urinary retention (AUR), need for surgery) but correlates poorly with symptoms. **Symptoms decide whether to operate; size and shape decide which operation.** A gland >30 cc triples AUR risk (Olmsted County).

2 · WORK-UP AND INDICATIONS FOR SURGERY

Work-up (AUA statements)

- History, examination, IPSS and urinalysis at baseline (St 1); repeat IPSS after treatment (St 3). IPSS bands: **0–7 mild, 8–19 moderate, 20–35 severe**.
- Before surgery: post-void residual (PVR) **required** (St 6); **consider** size/shape assessment by transrectal ultrasound (TRUS), abdominal ultrasound, cystoscopy, magnetic resonance imaging (MRI) or computed tomography (CT) (St 5), uroflowmetry (St 7), pressure–flow study if diagnosis uncertain (St 8).
- Cystoscopy shows a median lobe, stricture or tumour but is unreliable for size.
- Prostate cancer: shared decision on prostate-specific antigen (PSA) testing; an elevated PSA warrants biopsy **before** BPH surgery, because significant cancer needs radical treatment, not TURP.
- Counsel every patient about possible failure and retreatment (St 9).

Indications for surgery

AUA Statement 26 (surgery recommended)	† European Association of Urology (EAU) — absolute vs relative
Renal insufficiency secondary to BPH Refractory urinary retention Recurrent urinary tract infections (UTIs) Recurrent bladder stones or gross haematuria due to BPH LUTS refractory to, or patient unwilling to take, medical therapy	**Absolute** (complicated BPH): refractory retention, upper-tract dilatation/renal impairment, recurrent UTI, bladder stones, recurrent haematuria refractory to 5α-reductase inhibitor (5-ARI) **Relative**: moderate–severe bothersome LUTS failing or declining medical therapy

Peri-operative antibiotics (AUA Best Practice Statement 2019)

- Single dose suffices for most clean-contaminated transurethral cases; cover gram-negative enterics and enterococci (1st/2nd-generation cephalosporin or trimethoprim–sulfamethoxazole), guided by local antibiogram and pre-operative urine culture.
- **Active symptomatic UTI → defer surgery** until treated and asymptomatic.

3 · CHOOSING THE OPERATION: SIZE, SHAPE, BLEEDING RISK, SEXUAL FUNCTION

Prostate volume / situation	AUA 2023 options	Restrictions
Small, <30 cc	TUIP, TURP, TUVP, PVP, HoLEP, ThuLEP, TIPD	TUIP only ≤30 cc; TIPD 25–75 cc, no obstructing middle lobe
Average, 30–80 cc	TURP, TUVP, PVP, HoLEP, ThuLEP, PUL, WVTT, RWT, TIPD	PUL: no obstructing median lobe; WVTT and RWT treat median lobes
Large >80–150 cc / very large >150 cc	HoLEP, ThuLEP, simple prostatectomy	TURP, PVP, TUVP stop at 80 cc
Higher bleeding risk (cannot stop anticoagulation)	HoLEP, ThuLEP, PVP (St 43)	Otherwise hold anticoagulant/antiplatelet after liaising with prescriber
Wants erectile and ejaculatory preservation	PUL or WVTT (St 37)	Within 30–80 cc

KEY POINT Only enucleation spans every size band; only simple prostatectomy joins it above 80 cc. TUIP is the only option confined to small glands. MISTs and RWT live in the 30–80 cc band — so cystoscopic or imaging assessment of volume and median lobe precedes every recommendation.

4 · MONOPOLAR TURP (M-TURP) — THE BENCHMARK

- Electrified wire loop removes transition-zone (TZ) adenoma from bladder neck to verumontanum, down to the surgical capsule. AUA: TURP for <30 and 30–80 cc (St 28); monopolar or bipolar by surgeon expertise (St 29).
- Monopolar current passes loop → patient → grounding pad, so irrigant must be non-ionic (sterile water, **1.5% glycine**†, sorbitol) and is **hypo-osmolar** — the root of TUR syndrome.

Pre-operative set-up

- Spinal anaesthesia historically preferred: the awake patient shows early TUR syndrome (confusion, visual loss, nausea).
- Lithotomy; buttocks at table edge so the scope reaches the anterior prostate. Baseline abdominal examination.
- Grounding pad away from any joint prosthesis; irrigation at body temperature and lowest height giving vision.

Operative steps

ALGORITHM
Inspect bladder (tumour → resect it first, defer TURP; diverticula) → identify ureteric orifices, bladder neck, trigone, verumontanum, external sphincter (indigo carmine/methylene blue if orifices hidden) → resect median lobe first → lateral lobes → anterior tissue and apex last → evacuate chips (Ellik/Toomey) → final haemostasis → large-bore three-way catheter

- **Nesbit-type encirclement**: start at 11–9 and 1–3 o'clock, expose (not resect) bladder-neck fibres, carry to the verumontanum. **Floor-first**: channel at 5 or 7 o'clock to capsule (sets depth early) → widen laterally → lateral lobes "fall" into the fossa → smooth floor last.
- Long "canoe" chips; short chips leave an irregular bed that hides bleeders.
- **Why anterior and apex last**: anterior fossa has least adenoma and perforates easily; the sphincter is tilted (anterior part more proximal); the apex is best done in a bloodless field. Leaving

a little apical tissue is better than incontinence.

- Avoid repeated passes across the prostatovesical junction: over-resecting the dorsal bladder neck makes the scope go "uphill" and can undermine/detach the trigone.
- Catheter balloon volume matched to the cavity (so it does not fall into the fossa); brief traction. Continuous bladder irrigation (CBI) optional; unrecognised outflow block distends the bladder and worsens bleeding.

	Arterial bleeding	**Venous (sinus) bleeding**
Appearance	Bright red, persists during filling and draining	Darker; disappears with a full bladder (tamponade)
Detection	Bladder-neck (anterior) bleeders seen best with bladder emptied	Look with bladder empty and inflow off
Control	Always control: loop tamponade then precise fulguration; "ricochet" bleeders may need deeper resection to expose	Often not fully controllable → after arteries are secured, overfill balloon (50–60 mL) with traction

Post-operative care

- Even an uncomplicated TURP absorbs **800–1000 mL** irrigant — account for it in fluid prescription. Bladder spasm or rectal pain → check catheter for clot obstruction.
- Catheter out at 24–48 h; stool softeners 2–4 weeks; avoid perineal pressure (cycling, riding) 4–6 weeks; sex abstinence ≥2 weeks.
- Warn: dysuria while re-epithelialisation occurs (weeks); eschar passage with delayed bleeding at **1–4 weeks**; storage symptoms may persist — if emptying confirmed, antimuscarinic or β3-agonist; detrusor overactivity may take months to settle.
- Outcomes: large gains in IPSS, Qmax, quality of life (QoL) and PVR, durable ~10 years, low retreatment.

5 · TURP COMPLICATIONS AND THEIR MANAGEMENT

TUR syndrome

- **Mechanism**: prostatic venous pressure ≈ **10 mm Hg**; irrigation at higher pressure enters opened sinuses → dilutional hyponatraemia, hypervolaemia. Glycine adds ammonia toxicity (encephalopathy) and transient visual disturbance/blindness.
- **Features**: confusion, nausea, vomiting, visual change, hypertension then bradycardia, tachypnoea → seizures, coma, death.
- **Risk factors**: gland **>45 cc**, resection **>90 min**, capsular perforation, high irrigant height. Keep bag ≈ **60 cm** above patient; +10 cm more than doubles absorption.
- **Diagnosis**: neurological status + serum Na^+ (check after long/large resections). Na^+ **<120 mEq/L** = severe (coma, seizures, herniation).

ALGORITHM

Suspect TUR syndrome → stop/expedite surgery, achieve haemostasis, stop irrigation → send Na^+ → †loop diuretic if fluid-overloaded → severe central nervous system (CNS) symptoms: **3% saline 100 mL bolus** (raises Na^+ 2–3 mEq/L), repeat up to ×2 at 10-min intervals (max 300 mL); target rise 4–6 mEq/L; ≤12 mEq/L in first 24 h acceptable in acute post-TURP hyponatraemia (≤8–10 mEq/L/24 h if presentation >48 h) → high-dependency unit (HDU) monitoring

> **KEY POINT** Why is acute post-TURP hyponatraemia safer to correct than chronic? The brain has not yet shed its organic osmolytes, so osmotic demyelination (central pontine myelinolysis) is less likely. Bipolar TURP with saline eliminates TUR syndrome.

Other intra-operative problems

- **Perforation** (capsule, prostatovesical junction, bladder): sign = glistening fat. Usually extraperitoneal → limited: prolonged catheter drainage; extensive: percutaneous/open drainage.

Dome perforation → cystogram; intraperitoneal → laparotomy and repair. Doubt → cystogram with drainage films.

- **Ureteric orifice injury**: identify early (70° lens, IV dye); high bladder neck/large median lobe → resect median lobe from midline first, then orifices appear.
- **Intra-operative erection**: often subsides; else intracorporeal phenylephrine.

Late complications

Complication	Cause / features	Management / prevention
Bladder neck contracture (BNC)	Over-resection + excessive fulguration of bladder neck (commoner in small glands); undermined trigonal flap. Good early flow that declines; average onset ~6 months	Office cystoscopy → dilation → endoscopic incision (Collins knife, urethrotome or laser) over a wire until scar springs open, deep to perivesical fat (sphincter is distant after TURP) → recurrent: repeat incision + mitomycin C (10 mL of 0.4 mg/mL) → intractable: open/robotic Y-V plasty. Obliterated neck: IV methylene blue + suprapubic pressure to find the jet
Urethral stricture	Resectoscope trauma, infection; often meatus/fossa navicularis	Calibrate/dilate meatus 2 Fr above sheath, generous lubrication (30 mL gel), appropriate sheath size
Retrograde ejaculation	Bladder neck resected — about **two-thirds**	Counsel pre-operatively; erection, orgasm, desire unaffected
Incontinence	Internal sphincter lost at TURP, so continence depends on external sphincter; apical over-resection → total stress urinary incontinence (SUI). Rare, usually transient	Verumontanum = distal limit; †pelvic floor exercises; persistent SUI → †sling/artificial urinary sphincter
Bleeding	Early = incomplete haemostasis; delayed (1–4 weeks) with eschar	Return to theatre for early major bleeding; delayed mild → rest, fluids, †catheter/clot evacuation if retention
Erectile dysfunction (ED)	No conclusive evidence TURP causes it (neurovascular bundles stay near body temperature)	—
Retreatment	Residual apical/anterior tissue, gland >80 cc, regrowth, BNC, stricture	Rare; risk rises with size and pre-operative retention

TURP on anticoagulation carries significant bleeding risk → AUA prefers HoLEP, ThuLEP or PVP (St 43).

6 · BIPOLAR TURP (B-TURP)

- Active and return poles on the same loop: current stays at the resection site → **normal saline irrigation → no TUR syndrome**, so large resections and meticulous haemostasis are possible without time pressure.
- Plasma system: energy vaporises saline around the loop into plasma (orange glow of excited sodium ions); cuts at lower temperature while sealing vessels; platinum–iridium loop. Technique identical to m-TURP, often faster.

	Monopolar	Bipolar
Current path / irrigant	Through patient to pad / hypo-osmolar non-ionic	Local / isotonic saline
TUR syndrome	Yes (only procedure associated)	Eliminated (NNT ≈ 50)
Transfusion, clot retention (NNT ≈ 20), Hb drop	Higher	Lower
Catheter time, length of stay (LOS)	Longer	Shorter

	Monopolar	Bipolar
IPSS, QoL, PVR to 5 years; stricture, BNC, sexual function, retreatment	Equivalent / broadly similar	

b-TURP is equally effective and safer — likely to replace m-TURP as the gold standard.

7 · TRANSURETHRAL INCISION OF THE PROSTATE (TUIP)

- Concept: releases the capsule and bladder neck rather than removing adenoma. **Only <30 cc** (St 31) without a median lobe (relative contraindication; EAU same).
- Ideal patient: young man, small gland, concerned about ejaculation or fertility.

ALGORITHM
Collins knife / thin loop / holmium–thulium laser → incision at 5 and/or 7 o'clock, starting distal to ureteric orifice → through bladder neck → end proximal to verumontanum → deepen to capsule (± fat) → cautery → catheter out, discharge day 0–1

- vs TURP: shorter, less bleeding, similar symptom relief, smaller Qmax gain, **more reoperation**; less retrograde ejaculation; BNC not seen. May be displaced by MISTs.

8 · BIPOLAR TRANSURETHRAL VAPORISATION (TUVP)

- TURP equipment with a broad electrode (VaporTrode roller; bipolar PlasmaButton): higher energy over larger contact vaporises with simultaneous coagulation → excellent haemostasis and vision; no new capital; **no specimen**.
- Only **bipolar** TUVP recommended (AUA St 32: <30 and 30–80 cc). Monopolar abandoned: up to 10 mm coagulation → irritative symptoms, SUI, poor durability.
- vs TURP: shorter catheter time, fewer transfusions, essentially no perforation; but more dysuria, more recatheterisation for AUR and higher retreatment; retrograde ejaculation as common.

> **KEY POINT** Vaporisation trades haemostasis for a coagulated, specimen-free bed: the coagulum explains dysuria and retention; the lack of a clean anatomical plane explains weaker durability. The same trade recurs with PVP.

9 · LASER PHYSICS

- Coagulation (~60 °C, delayed sloughing) abandoned for vaporisation (≥100 °C, immediate). Effect depends on wavelength and chromophore; deeper penetration = more coagulum.

Laser	Wavelength	Chromophore	Penetration	BPH use
potassium titanyl phosphate (KTP) / lithium triborate (LBO) "GreenLight"	532 nm	Haemoglobin	0.8 mm	PVP
Holmium:YAG (pulsed)	≈2120 nm	Water	0.4 mm	HoLEP, TUIP
Thulium:YAG / thulium fibre laser (TFL)	2013 / 1940 nm	Water (TFL at absorption peak; ~4× holmium)	**0.2 mm** (shallowest)	ThuLEP, ThuVEP, ThuVARP
Diode	† 940–1470 nm	† Water and haemoglobin	0.5–5 mm	Vaporisation/enucleation; needs comparative data
Nd:YAG	1064 nm	—	≈1 cm	Abandoned (sloughing, retention)

Laser safety: wavelength-specific eyewear for all (including patient), warning signage, covered windows, trained operators.

10 · PHOTOSELECTIVE VAPORISATION OF THE PROSTATE (PVP, GREENLIGHT)

- Frequency-doubling Nd:YAG through a KTP/LBO crystal halves wavelength to 532 nm → absorbed by haemoglobin, not water → efficient vaporisation with a thin coagulation rim → haemostasis; sinuses sealed, saline irrigation, no TUR syndrome.
- 120 W or 180 W platforms. AUA: 120/180 W PVP for <30 and 30–80 cc (St 33) and **higher bleeding risk** (St 43). Not >80 cc (high retreatment).

Technique

- 23–24 Fr cystoscope, side-firing fibre (70°). Start at bladder neck: identify orifices → open neck with midline or 5/7 o'clock incisions → vaporise posterior neck level with trigone. Aim laterally, never at orifices; avoid heavy coagulation at the neck (BNC).
- Sweep 15–30° (novice: 30° in 1–2 s), constant working distance (too close damages fibre; too far coagulates). Large bubbles = efficient vaporisation; no bubbles = charring → more dysuria. Vaporise posterior tissue in horizontal layers, never aim at the rectum.
- Bleeding (apex, median lobe, bladder neck, posterolateral): moderate → vaporise around feeder; pulsatile → coagulation mode ~30 W; refractory → Bugbee (non-ionic irrigant) or convert to resection.
- Post-op: catheter out day 0–1; tissue passage at 7–10 days. Energy >5 kJ/cc marks more complete vaporisation.

Complications and management

- **Dysuria/storage symptoms** — highest of transurethral methods, proportional to coagulated volume; self-limiting ≤3 months; non-steroidal anti-inflammatory drugs (NSAIDs) or short steroid taper; exclude tissue flap or retained fibre fragment.
- **Infection** (UTI, epididymitis) — necrotic coagulum harbours bacteria; culture-directed, longer antibiotic courses.
- Transfusion very rare even on anticoagulants; capsular perforation 0.2–1%; stricture/BNC similar to TURP; retrograde ejaculation similar to TURP; erectile function minimally affected.
- Anticoagulated men improve equally without meaningful rise in transfusion.

11 · HOLMIUM LASER ENUCLEATION OF THE PROSTATE (HOLEP)

- Anatomic endoscopic enucleation of the prostate (AEEP): the adenoma is shelled out along the capsular plane, endoscopic equivalent of open simple prostatectomy. "Enucleation is enucleation" — the plane matters more than the energy.
- Pulsed Ho:YAG, 0.4 mm penetration, coagulates vessels within 2–3 mm. Evolution: HoLAP → HoLRP → HoLEP (1998; made practical by the tissue morcellator).
- **Size-independent**: <30, 30–80 and >80 cc, with or without median lobe (St 38); suitable for higher bleeding risk (St 43). Only endoscopic procedure with efficacy **superior** to TURP.
- Steep learning curve: ≥20 supervised cases; efficiency plateau ≥50 cases.

Technique (three-lobe, bottom-up, early apical release)

ALGORITHM

550 µm end-firing fibre, 24–28 Fr continuous-flow resectoscope, saline, ~2 J × 40–50 Hz (coag 1–2 J × 20 Hz) → 5 and 7 o'clock bladder-neck incisions to capsule (6 o'clock single incision if no median lobe) → carry to just proximal to verumontanum, join, enucleate median lobe into bladder → perimontanal inverted-U incision, develop lateral lobe–capsule plane at apex → sweep anteriorly to 11–12 and 12–1 o'clock → incise apical mucosal strip (avoids sphincter traction) → anterior plane to bladder neck ("Baleen" sign heralds bladder) → release lobes into bladder → haemostasis (defocused beam) → morcellation via nephroscope → 22 Fr three-way catheter, ~60 mL balloon

- The smooth, reflective capsule is the depth reference: too shallow leaves adenoma and bleeding feeders; too deep perforates (heals with a few days of catheter).

- **Morcellation safety**: two fully open inflow lines, distended bladder, haemostasis first. If mucosa engaged → stop cutting and suction, disengage. Superficial injury → catheter; intraperitoneal → laparotomy and repair.
- Retained adenoma does not dissolve → irritative symptoms, bleeding, infection → re-morcellate.

Complications

- **Transient incontinence** common early (as after open prostatectomy — reflects volume removed), rarely persistent; early apical release reduces it. Splitting the commissure carelessly can cause permanent SUI.
- **BNC** uncommon, concentrated in small glands (<40 cc → consider prophylactic bladder-neck incision). Stricture uncommon (meatal; long operation/catheter).
- Retrograde ejaculation ≈ **three-quarters** (whole TZ removed); median-lobe-only HoLEP preserves it in ~90%. Erectile function unchanged or improved.
- Retreatment rare (whole TZ removed — negligible regrowth vs TURP, which leaves a peripheral rim). Durable to 18 years.
- On anticoagulants: slightly longer LOS and CBI; warfarin fares worse than direct oral anticoagulants.

12 · THULIUM ENUCLEATION (THULEP) AND VARIANTS

- Tm:YAG 2013 nm (continuous or pulsed) and TFL 1940 nm (water absorption peak) → clean, precise cut, 0.2 mm penetration; continuous wave may carbonise.
- Variants: vaporesection (ThuVARP), vaporisation (ThuVAP), vapoenucleation (ThuVEP — laser "vapo-incision"), ThuLEP (mainly blunt dissection; laser for mucosal incision, lobe division and haemostasis).
- Same steps and morcellation as HoLEP.
- ≈ HoLEP in function; less bleeding than TURP. St 38 and 43. Retrograde ejaculation ~four-fifths; erections preserved.

	HoLEP	ThuLEP / ThuVEP
Wavelength / mode	≈2120 nm pulsed	2013 nm continuous/pulsed; TFL 1940 nm
Penetration	0.4 mm	0.2 mm
Learning curve	Steep (plateau ≥50)	Claimed shorter (~30)
Evidence	Largest; 18-year follow-up	Shorter follow-up

13 · MINIMALLY INVASIVE SURGICAL THERAPIES (MISTS)

Ideal minimally invasive surgical therapy (MIST): office-based, single session, no general/regional anaesthesia, better than drugs short-term, avoids the sexual side effects and recovery of surgery, removes lifelong medication. **Trade-off**: retreatment within 5 years exceeds surgery and durability beyond 5 years is uncertain — warn patients.

Prostatic urethral lift (PUL, UroLift)

- Non-ablative: permanent transprostatic implants compress lateral lobes to open the prostatic urethra. Implant: nitinol capsular tab + stainless-steel urethral end joined by a tensioned suture.
- Local anaesthesia; typically **4–6 implants** anterolaterally (between dorsal veins and neurovascular bundles), bladder neck to apex. Gland tissue is more compliant than capsule, so tension opens the lumen.
- AUA: **30–80 cc, no obstructing median lobe** (confirm by cystoscopy/ultrasound). US Food and Drug Administration (FDA) ≤100 cc not adopted by AUA.
- **L.I.F.T.** (vs sham): durable 5 years, mostly catheter-free, **no new sexual or ejaculatory dysfunction**. **BPH6** (vs TURP): antegrade ejaculation 100% vs ~61%, faster recovery; TURP better for IPSS, Qmax, PVR and retreatment.

- Complications: mild dysuria, haematuria, urgency <2 weeks; **encrustation** of implants protruding into bladder → deploy >1.5 cm distal to bladder neck; remove with endoscopic graspers.

Water vapour thermal therapy (WVTT, Rezūm)

- **Convective** energy: radiofrequency generates steam injected into the TZ; condensation releases latent heat (540 cal/g) → necrosis in a 1.5–2 cm lesion. Prostatic collagen (capsule) confines heat to the TZ. Only ~4 kJ per procedure.
- Rigid scope, 18-gauge needle; 9-s injections at 103 °C, first 1 cm distal to bladder neck at 3 and 9 o'clock, then every 1 cm to proximal verumontanum; extra injections for **median lobe (treatable)**. Oral sedation/prostatic block; ~5 min.
- Rectal safety: never aim straight posteriorly near verumontanum — angle 45° off midline.
- **Rezūm II** (vs sham): ~50% IPSS improvement durable to 5 years; surgical retreatment 4.4% (underestimated — high loss to follow-up); **no de novo ED**; transient reduced ejaculate settles by 12 months.
- Catheter for a few days; irritative symptoms up to ~3 weeks. AUA St 36/37.

> **KEY POINT** PUL moves tissue; WVTT ablates it. That is why PUL cannot treat a median lobe, needs no catheter and causes minimal dysuria, whereas WVTT treats median lobes at the cost of a catheter and weeks of irritative symptoms. Both preserve the bladder neck and hence antegrade ejaculation.

14 · ROBOTIC WATERJET TREATMENT (RWT, AQUABLATION)

- Non-thermal, high-velocity saline waterjet; surgeon maps resection on real-time biplanar TRUS; robot ablates in minutes (depth up to 2.4 cm). Bladder neck, verumontanum and sphincter excluded from plan. **Needs general/spinal anaesthesia — not a MIST**. AUA 30–80 cc (St 39); median lobe treatable; no specimen; mainly posterior (4–8 o'clock) tissue.
- **Haemostasis is the weakness** (no cautery): originally catheter traction → bleeding/transfusion; now **focal bladder-neck cautery** with a resectoscope after clearing fluffy remnants.
- **WATER** (vs TURP): non-inferior, durable to 5 years; retrograde ejaculation 10% vs 36% (likely underestimated).
- Rectal injury: rare, catastrophic.

15 · TRUE MISTS AND TEMPORARY DEVICES

- "True" MISTs: flexible cystoscopy, no sedation, preserve sexual function; deliberately limited durability (12–18 months target).
- **iTind (TIPD)**: three nitinol struts (12, 5, 7 o'clock) + 6 o'clock anchoring leaflet; left **5–7 days** → ischaemic pressure necrosis remodels bladder neck and prostatic urethra; removed by collapsing into an 18 Fr catheter. AUA St 41: 25–75 cc, no obstructing median lobe. Sham RCT: responder end point met, absolute IPSS difference not significant; sexual function preserved.
- **Drug-coated balloon (Optilume)**: dilation with anterior commissurotomy + paclitaxel balloon (≥5 min) to prevent re-scarring; single-arm data only; not in AUA list.

16 · PROSTATE ARTERY EMBOLISATION (PAE)

ALGORITHM
Femoral access under local anaesthesia → pelvic angiography → catheter balloon with contrast as landmark → 1.9–2.4 Fr microcatheter into prostatic arteries (most often from internal pudendal; often duplicated) → microspheres until stasis — bilateral better than unilateral

- **Contraindications**: underactive/acontractile or neurogenic bladder, stricture/BNC, large diverticula or stones, contrast allergy, chronic kidney disease. Atherosclerosis/tortuosity may prevent completion; high radiation.

- AUA 2023 St 40: may be offered by trained clinicians; no size limit; role in refractory BPH haematuria. EAU: for men accepting inferior results to TURP.
- vs TURP: inferior in IPSS, Qmax and volume reduction; shorter stay. Poor durability — majority reintervene by 10 years.
- Complications: access-site haematoma/pseudoaneurysm; non-target embolisation (bladder, rectal, penile ischaemia — usually conservative); post-embolisation syndrome, AUR. Ejaculation and erection largely preserved.

17 · OVERALL COMPARISON

Procedure	Volume (AUA)	Anaesthesia / catheter	Efficacy vs TURP	Antegrade ejaculation	Key hazard	Retreatment
m-/b-TURP	<80 cc	Spinal/general anaesthesia (GA); 1–2 days	Benchmark	Lost ~⅔	Bleeding, TUR syndrome (mono)	Low
TUIP	<30 cc	Short; day 0–1	Slightly less	Mostly kept	—	Higher
b-TUVP	<80 cc	Short catheter	Similar early, less durable	Lost	Dysuria, AUR	Higher
PVP	<80 cc; anticoagulated	Day 0–1	Non-inferior (GOLIATH)	Lost ~⅔	Dysuria, UTI	Higher than TURP
HoLEP/ThuLEP	Any; anticoagulated	GA/spinal; day 0–1	**Superior** (HoLEP)	Lost ~¾–⅘	Transient SUI, morcellator injury	Lowest
RWT	30–80 cc	GA/spinal	Non-inferior	Mostly kept	Bleeding	Low (short data)
PUL	30–80 cc, no median lobe	Local; often none	Less	**Preserved**	Encrustation	~14% at 5 y
WVTT	30–80 cc	Sedation; few days	Less	**Preserved**	Irritative symptoms	~4–8% at 5 y
iTind	25–75 cc, no median lobe	Local; device 5–7 d	Uncertain	**Preserved**	—	Unknown
PAE	Any	Local; day case	Inferior	Largely kept	Non-target embolisation	High

KEY POINT The "gold standard" for a given man is the operation that meets his goals safely — chosen by shared decision-making around his most bothersome symptoms, prostate size/shape, bleeding risk, wish to preserve ejaculation, and the surgeon's own expertise.

18 · LANDMARK TRIALS

Trial	Design	Result	Take-home
L.I.F.T.	PUL vs sham, 2:1, 30–80 cc	IPSS ~halved; 13.6% surgical retreatment at 5 y	PUL works, preserves sexual function
BPH6	PUL vs TURP, ≤60 cc	Composite favoured PUL; TURP better IPSS/Qmax/PVR	"Superior" refers only to composite
Rezūm II	WVTT vs sham, 30–80 cc incl. median lobe	IPSS −11.2 vs −4.3; durable 5 y; no de novo ED	Convective MIST incl. median lobe
WATER	RWT vs TURP, double-blind	Non-inferior; better in 50–80 cc; less retrograde ejaculation	Aquablation for 30–80 cc

Trial	Design	Result	Take-home
GOLIATH	180 W PVP vs TURP, ≤100 cc	Non-inferior to 2 y; shorter catheter and stay	PVP alternative to TURP
UK-ROPE	PAE vs TURP, matched registry	PAE inferior for IPSS/QoL; shorter stay	PAE is a lesser-efficacy option

19 · MCQ ONE-LINERS

1. Only procedure associated with TUR syndrome: monopolar TURP.
2. TUR syndrome risk factors: gland >45 cc, resection >90 min.
3. Prostatic venous pressure ≈ 10 mm Hg; irrigant height ≈ 60 cm.
4. 3% saline 100 mL raises sodium by 2–3 mEq/L; maximum 300 mL.
5. Retrograde ejaculation after TURP ≈ two-thirds; after HoLEP ≈ three-quarters.
6. Distal landmark protecting the external sphincter at TURP: verumontanum.
7. Recurrent BNC: transurethral incision plus mitomycin C; intractable: Y-V plasty.
8. TUIP: only glands <30 cc without median lobe.
9. Only endoscopic procedure with efficacy superior to TURP: HoLEP.
0. HoLEP and ThuLEP are size-independent (AUA St 38).
11. Higher bleeding risk: HoLEP, ThuLEP or PVP (St 43).
12. PVP wavelength 532 nm, chromophore haemoglobin.
13. Nd:YAG abandoned: ~1 cm penetration causing sloughing and retention.
14. Highest dysuria among transurethral techniques: PVP.
15. PUL contraindication: obstructing median lobe.
16. Avoid PUL encrustation: implants >1.5 cm distal to bladder neck.
17. WVTT energy: convection, 103 °C steam for 9 s, ~4 kJ total.
18. WVTT median lobe needle: 45° off midline, never straight posterior.
19. Aquablation needs general/spinal anaesthesia, so is not a MIST.
20. iTind left in situ 5–7 days; AUA range 25–75 cc.
21. BNC after HoLEP is commoner in small glands (<40 cc).

152 Simple Prostatectomy: Open, Laparoscopic and Robotic

Reading time ≈ 19 min (slow pace)

1 · DEFINITION AND SURGICAL ANATOMY

- **Simple prostatectomy** = enucleation of the hyperplastic adenoma only, for benign prostatic hyperplasia (BPH); radical prostatectomy removes the whole gland for cancer. Gold standard for bladder outlet obstruction with glands **>80 g**.
- BPH arises in the transition zone (TZ); the enlarging TZ compresses the peripheral zone into the **surgical capsule (pseudocapsule)** — the cleavage plane → *Ch 148*.
- **Arterial supply**: prostatovesical artery (usually gluteopudendal trunk of internal iliac) → inferior vesical + prostatic arteries. **Urethral branches enter posterolaterally just distal to the bladder neck** and supply the adenoma — hence the universal **5 and 7 o'clock** figure-of-eight suture. Veins: dorsal venous complex (DVC) → internal iliac.
- Eponyms MCQ: **Freyer** popularised suprapubic transvesical; **Millin (1945)** retropubic; first laparoscopic 2002, first robotic 2008.

2 · INDICATIONS

- BPH, gland **>80–100 g**, bothersome lower urinary tract symptoms (LUTS) refractory to medical or prior endoscopic therapy.
- Complications of obstruction: urinary retention, recurrent urinary tract infection (UTI), **bladder stones**, recurrent prostatic haematuria, obstructive nephropathy.
- Concomitant **bladder diverticulum** needing surgery (relative); large calculi.
- Cannot be placed in lithotomy for transurethral resection of prostate (TURP).

3 · PRE-OPERATIVE EVALUATION

- History: International Prostate Symptom Score (IPSS), bleeding history/anticoagulants, prior pelvic or prostate surgery (adhesions dictate route).
- Digital rectal examination (DRE): size, nodules.
- Laboratory: prostate-specific antigen (PSA), renal function, complete blood count, coagulation, **urine culture** (treat before surgery).
- **Transrectal ultrasound (TRUS)**: accurate volume justifies enucleation over a less invasive option.
- Cancer suspicion → magnetic resonance imaging (MRI) ± biopsy. Cystoscopy → stones, diverticula, median lobe. **Urodynamics** if long-standing retention (atonic bladder?).
- General anaesthesia preferred; spinal acceptable for open only.

> **KEY POINT** Work-up answers three questions: is the gland big enough (TRUS)? is there cancer that makes a simple operation wrong (DRE, PSA, MRI)? will the bladder empty once unobstructed (urodynamics)?

4 · OPEN TECHNIQUES

22-Fr (French) urethral catheter; lower midline extraperitoneal incision → space of Retzius.

	Retropubic (Millin)	Suprapubic transvesical (Freyer)
Entry	Transverse **capsulotomy ≈2 cm distal to bladder neck** between stay sutures; bladder not opened	Transverse **cystotomy 1–2 cm proximal to bladder neck**; circumferential bladder neck mucosal incision, avoiding ureteric orifices

	Retropubic (Millin)	Suprapubic transvesical (Freyer)
Preliminary	Endopelvic fascia, puboprostatic ligaments divided; superficial DVC branch cauterised	Identify ureteric orifices
Enucleation	Metzenbaum scissors find plane → finger enucleation laterally/posteriorly; anterior commissure split; urethra divided **under vision** at apex proximal to sphincter; median lobe via bladder neck	Finger in urethra fractures anterior commissure (or posterior midline plane first) → circumferential to apex → urethra pinched between thumb and finger
Advantages	Direct view of adenoma and fossa; precise apical transection (sphincter safe); no bladder trauma	Treats large **median lobe, diverticula, bladder stones** simultaneously
Disadvantages	DVC bleeding; no bladder access	Bladder opened; apex blind; poorer fossa view
Closure	Capsule: two running 2-0 sutures, watertight; retropubic drain	Cystotomy two layers; drain; ± Malecot suprapubic tube (SPT)

- **Why transverse incisions?** A longitudinal capsulotomy/cystotomy may tear distally during retraction and injure the external sphincter.
- **Vesicocapsular** (vertical incision across bladder neck and capsule): easier to some, but more leak and sphincter risk.
- Catheter: 22-Fr three-way, 30-mL balloon (inflated to 50 mL so it cannot slip into the fossa).

KEY POINT All open variants = four moves: enter → pseudocapsular plane → divide urethra at apex **proximal to the verumontanum**, no traction on sphincter → 5/7 o'clock haemostasis + bladder neck mucosal advancement.

5 · HAEMOSTASIS

Technique	When / how
Pack fossa 5 min	First, for brisk bleeding
5 and 7 o'clock figure-of-eight (2-0 chromic/polyglactin)	Every case; deep through proximal capsule → ligates urethral arteries
Bladder neck mucosal advancement ("trigonisation")	Mucosa sutured into fossa over bare detrusor → haemostasis, prevents bladder neck contracture; avoid ureteric orifices
11 and 1 o'clock sutures	Anterior arterial branches, occasionally
Walsh prophylactic control	Before Millin: ligate DVC at apex and lateral pedicles, sparing neurovascular bundles
O'Conor capsular plication	Diffuse bleeding: transverse plicating sutures approximate posterior capsule
Malament purse-string	Persistent bleeding: nylon purse-string round bladder neck tied over catheter, brought out suprapubically over a button → partitions fossa from bladder; removed day 2–3

ALGORITHM
Postoperative bleeding → catheter traction (50 mL balloon) ± continuous bladder irrigation (CBI) → exclude coagulopathy → theatre: endoscopic clot evacuation + fulguration → reopen fossa if severe

6 · POSTOPERATIVE CARE (OPEN)

- Traction/CBI as needed; compression stockings; early ambulation; non-opioid analgesia.
- Day 2, clear urine: remove Foley, clamp SPT for trial of void. Pelvic drain out if low; high output → **drain creatinine** (leak).
- No heavy lifting ≈4 weeks; review at 6 weeks and 3 months (symptoms, postvoid residual [PVR]).

7 · LAPAROSCOPIC AND ROBOTIC SIMPLE PROSTATECTOMY

- Laparoscopic (LSP): less pain and blood loss than open but steep learning curve, poor ergonomics — largely supplanted by robotic-assisted simple prostatectomy (RASP).
- **Multiport RASP**: usually transperitoneal (extraperitoneal for hostile abdomen); replicates suprapubic or Millin steps.

ALGORITHM
Vertical cystotomy, stay sutures → identify orifices → bladder neck incision 3→9 o'clock posteriorly under median lobe → develop **posterior plane** (most distinct) to apex → circumferential lobes → divide anterior commissure, see verumontanum → transect urethra **just proximal to veru** → 5/7 o'clock barbed sutures → **vesicourethral mucosal advancement** (double-armed suture from 6 o'clock) → two-layer bladder closure

- Avoid posterolateral cautery — thermal neurovascular bundle injury → erectile dysfunction.
- Mucosal advancement re-covers the fossa → usually **no CBI, drain or SPT**; home day 1, catheter out day 5–7.
- **Single-port transvesical (STEP)**: 2.7-cm suprapubic incision, no peritoneal entry, **pneumovesicum 8 mm Hg** → same-day discharge, catheter out day 2. Competes with laser enucleation for large glands.

8 · OUTCOMES AND COMPARISONS

Comparison	Bottom line
Robotic vs open	Robotic: less blood loss/transfusion, shorter catheter and stay, fewer complications; longer operating time. Functional outcomes (IPSS, maximum flow rate [Qmax], PVR) **equal**
Single-port vs multiport	Shorter stay, less CBI and opioid; same complications
vs TURP	Lower retreatment, complete removal under vision, **no transurethral resection (TUR) syndrome**, treats stones/diverticula; open: incision, longer stay, more bleeding
†vs holmium laser enucleation (HoLEP)	†Randomised trials in large glands: equivalent durable symptom and flow improvement; HoLEP less bleeding, shorter catheter/stay, but long learning curve and cannot treat diverticula. Size-independent endoscopic alternative → *Ch 151*

KEY POINT Minimally invasive routes improve perioperative morbidity only; function is identical because the enucleation is identical. Open transfusion now **<5%**.

9 · COMPLICATIONS AND MANAGEMENT

Complication	Cause / rate	Management
Haemorrhage	Urethral arteries, DVC	Escalation algorithm above
Urinary extravasation	Incomplete capsule/cystotomy closure; unrecognised capsular tear	High drain output, drain creatinine = urine → keep drain + Foley; resolves
Ureteric orifice injury	Bladder neck incision, haemostatic or advancement sutures	Avoid; stent if orifices close
Bladder neck contracture	Open **2–6%**; minimally invasive <1%	Dilation or Collings knife incision to 22 Fr
Storage LUTS	Weeks–months	Anticholinergic or β3-agonist
Stress incontinence	Rare — sphincter untouched	Mild: bulking injection; severe: artificial urinary sphincter
Erectile dysfunction	**3–5%**	—
Retrograde ejaculation	**80–90%**	Counsel pre-op

Complication	Cause / rate	Management
UTI, urethral stricture, venous thromboembolism	Uncommon; thromboembolism <1%	Culture-directed antibiotics; prophylaxis

KEY POINT Why? The adenoma lies inside the pseudocapsule and proximal to the verumontanum: removing it destroys the bladder neck (retrograde ejaculation) but spares the external sphincter and extracapsular neurovascular bundles (incontinence, erectile dysfunction rare).

10 · MCQ ONE-LINERS

1. Simple prostatectomy threshold: gland >80 g.
2. Enucleation plane: surgical (pseudo)capsule = compressed peripheral zone.
3. Adenoma arteries enter posterolaterally at bladder neck: 5 and 7 o'clock.
4. Retropubic approach without opening bladder: Millin, 1945.
5. Suprapubic transvesical prostatectomy popularised by Freyer.
6. Bladder stones, diverticulum or large median lobe → suprapubic.
7. Millin capsulotomy: transverse, ≈2 cm distal to bladder neck.
8. Suprapubic cystotomy: 1–2 cm proximal to bladder neck.
9. Transverse incision avoids distal extension into sphincter.
0. Main Millin disadvantage: DVC bleeding.
11. Urethra divided proximal to verumontanum.
12. Catheter balloon 50 mL prevents migration into fossa.
13. Malament: purse-string partitioning fossa from bladder.
14. O'Conor: posterior capsular plication.
15. Suspected leak: check drain creatinine.
16. First laparoscopic simple prostatectomy: Mariano, 2002.
17. First robotic simple prostatectomy: Sotelo, 2008.
18. Robotic step removing need for CBI: vesicourethral mucosal advancement.
19. STEP uses pneumovesicum at 8 mm Hg.
20. Functional outcomes equal across open and robotic.
21. Retrograde ejaculation after enucleation: 80–90%.
22. Bladder neck contracture: incise with Collings knife to 22 Fr.
23. No TUR syndrome with simple prostatectomy.

★ Putting it together: the BPH pathway

Reading time ≈ 5 min (slow pace) · Integrates Ch 148–152 · use these as the skeleton of long-essay answers

A · MAN WITH LOWER URINARY TRACT SYMPTOMS (LUTS)

ALGORITHM

1. Assess: history and International Prostate Symptom Score (IPSS: 0–7 mild, 8–19 moderate, 20–35 severe), digital rectal examination, urinalysis, prostate-specific antigen (PSA) if it changes management, frequency–volume chart if nocturia. Uroflow and post-void residual (PVR) before invasive therapy (→ *Ch 150*).

2. Look for complications that make surgery the answer: refractory retention, recurrent urinary tract infection, bladder stones, recurrent haematuria from the prostate, renal impairment from obstruction.

3. Mild or not bothered → watchful waiting with lifestyle advice.

4. Bothersome → medical therapy (table B); if it fails or the patient prefers → procedure (table C).

B · MEDICAL THERAPY

Drug	Use	Key points
α1-blocker (tamsulosin, alfuzosin, silodosin)	First line for bothersome LUTS; works within days	Dizziness, ejaculatory dysfunction; intraoperative floppy iris syndrome — warn before cataract surgery
5α-reductase inhibitor (finasteride, dutasteride)	Enlarged prostate (>30–40 mL), long-term treatment intended	Slow onset (months); reduces retention and need for surgery; lowers PSA by 40–50% — double the measured value
Combination α-blocker + 5-ARI	Large gland, high progression risk	Best prevention of progression (MTOPS, CombAT)
Antimuscarinic or β3-agonist add-on	Persistent storage symptoms	Check PVR first
Tadalafil 5 mg daily	LUTS with erectile dysfunction	No nitrates
Desmopressin	Nocturnal polyuria	Monitor sodium, especially if ≥65

C · CHOOSING A PROCEDURE

Situation	Options
Small gland (<30 mL)	Transurethral incision (TUIP) or TURP
30–80 mL	TURP (bipolar), photoselective vaporisation (PVP), HoLEP/ThuLEP, Aquablation; MISTs (prostatic urethral lift, water vapour therapy)
>80 mL	HoLEP/ThuLEP (size-independent), open/robotic simple prostatectomy (especially with bladder stones or diverticulum) (→ *Ch 152*)
Cannot stop anticoagulation	HoLEP, ThuLEP, PVP
Wants ejaculation preserved	Prostatic urethral lift or water vapour therapy (no median lobe for urethral lift)
Unfit for anaesthesia	Long-term catheter or † prostatic artery embolisation

D · TURP COMPLICATIONS AT A GLANCE

KEY POINT TUR syndrome (monopolar, hypotonic glycine absorption): confusion, visual change, hyponatraemia → stop, check sodium, diuretic for overload, 3% saline for severe neurological symptoms, correct slowly. Bipolar saline resection prevents it. Late problems: retrograde ejaculation (about two-thirds), urethral stricture (meatus, bulb), bladder neck contracture (small glands, over-resection; ~6 months), incontinence (pelvic floor training, then sling or artificial sphincter) (→ *Ch 151*).